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Yoshimura et al.

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(54) **WORK VEHICLE HAVING A CONTROLLER THAT CONTROLS THE HYDRAULIC CYLINDERS AND THE ELECTRICALLY DRIVEN MOTOR ON A BASIS OF A DISTRIBUTION RATIO OF A FIRST TORQUE AND A DISTRIBUTION RATIO OF A SECOND TORQUE**

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(58) **Field of Classification Search**
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See application file for complete search history.

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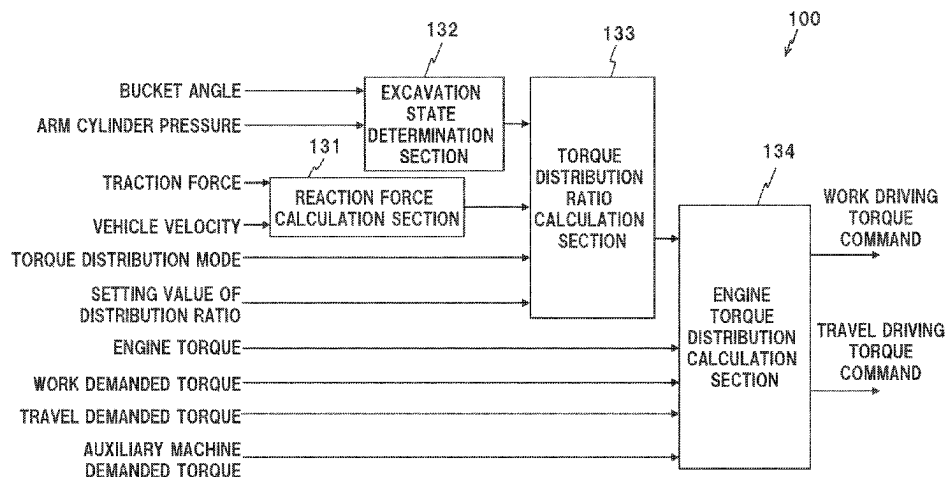
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(57) **ABSTRACT**

A work vehicle includes: an engine; a hydraulic pump that is driven by the engine; hydraulic cylinders that are extended and contracted by pressurized fluid delivered from the hydraulic pump; a work device that is moved according to the extension/contraction operations of the hydraulic cylinders

(Continued)



ders; a travel device that is driven independently of the work device; an electrically driven motor that is driven by electric power generated by the engine to operate the travel device; and a controller that controls the hydraulic cylinders and the electrically driven motor. The controller controls the output power of the hydraulic pump and the output power of the electrically driven motor by changing the distribution ratios of a first torque consumed by the work device and a second torque consumed by the travel device among torques output by the engine, on the basis of a reaction force received by the vehicle body.

5 Claims, 14 Drawing Sheets

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E02F 9/08 (2006.01)
F16H 59/14 (2006.01)

(52) U.S. Cl.

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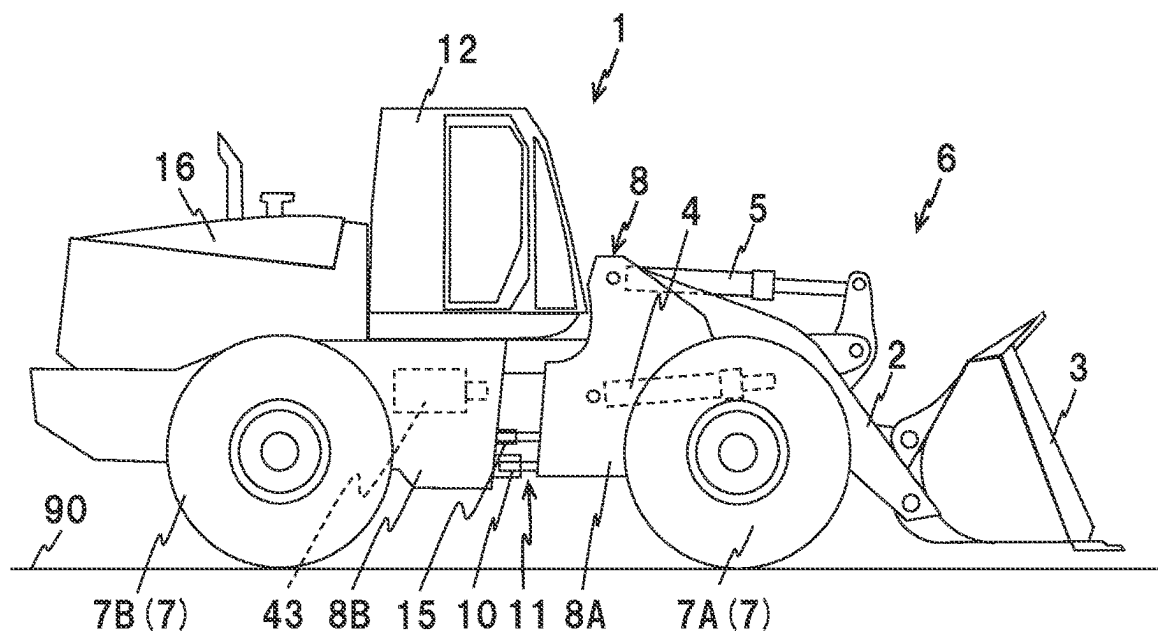
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FIG. 1



FILE

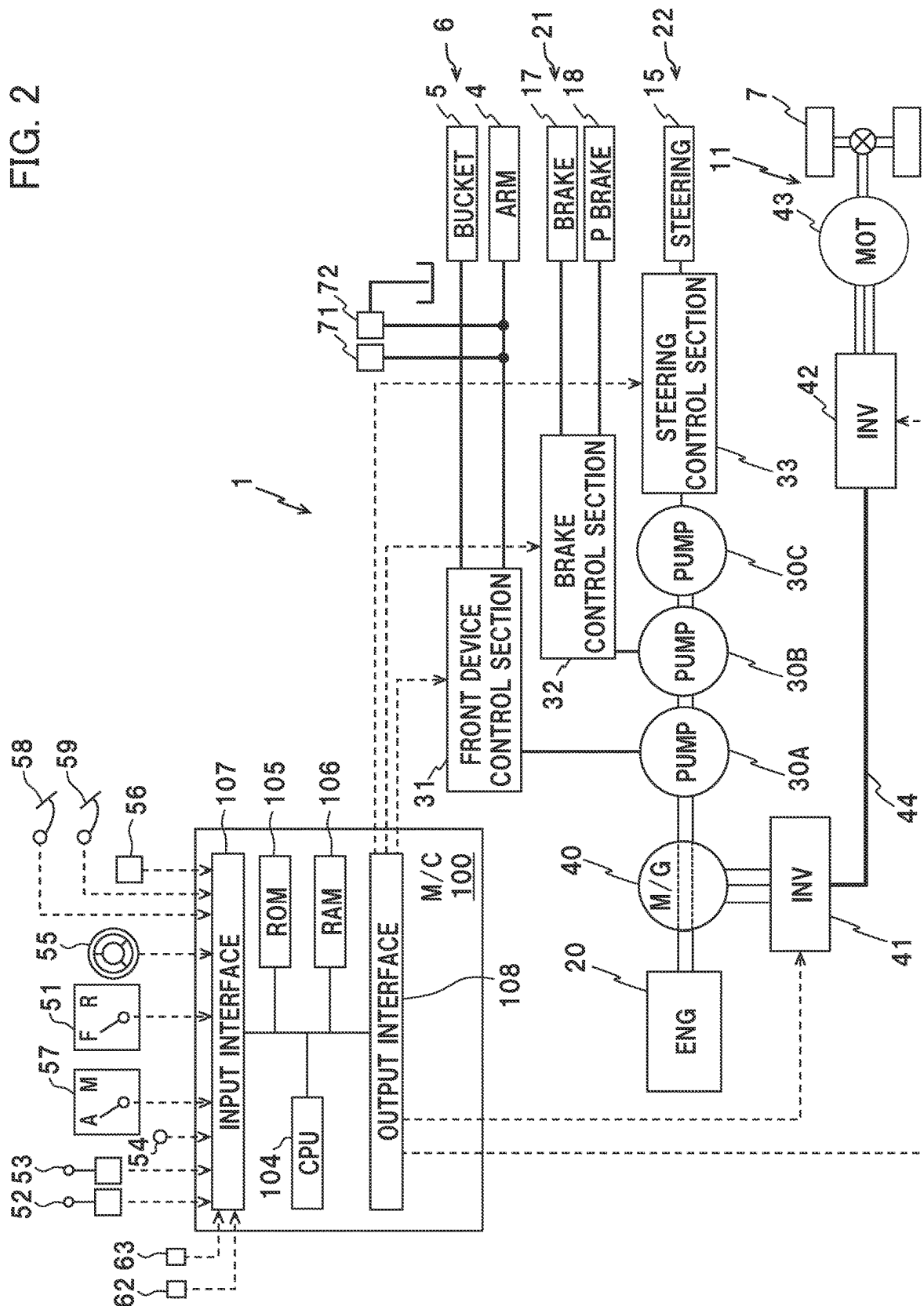


FIG. 3

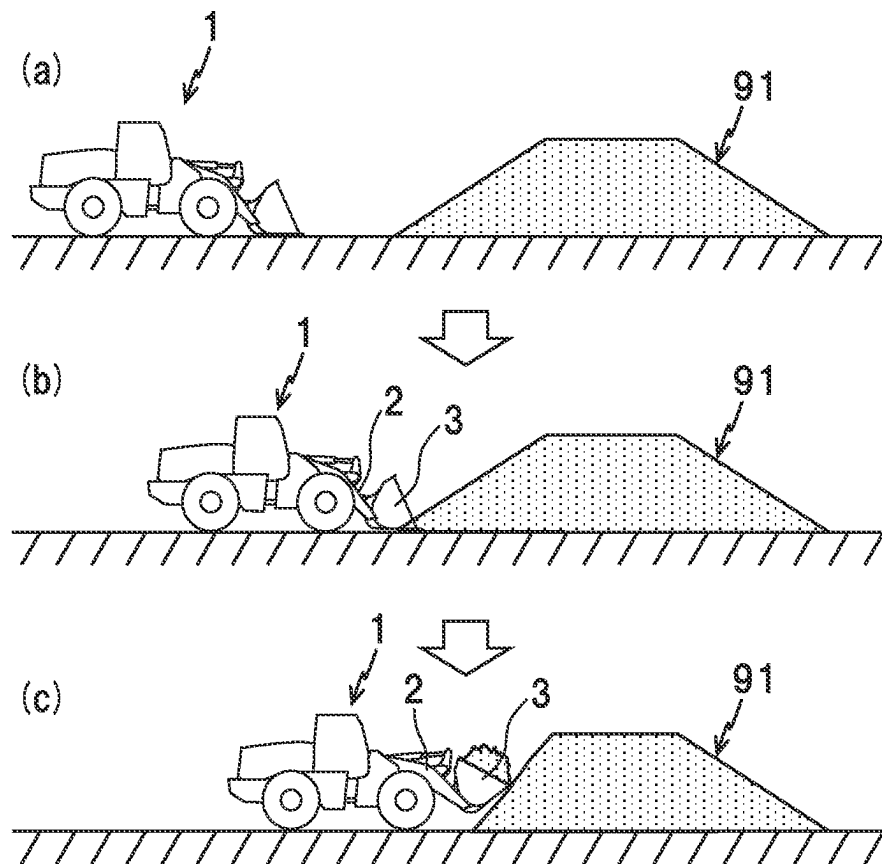


FIG. 4

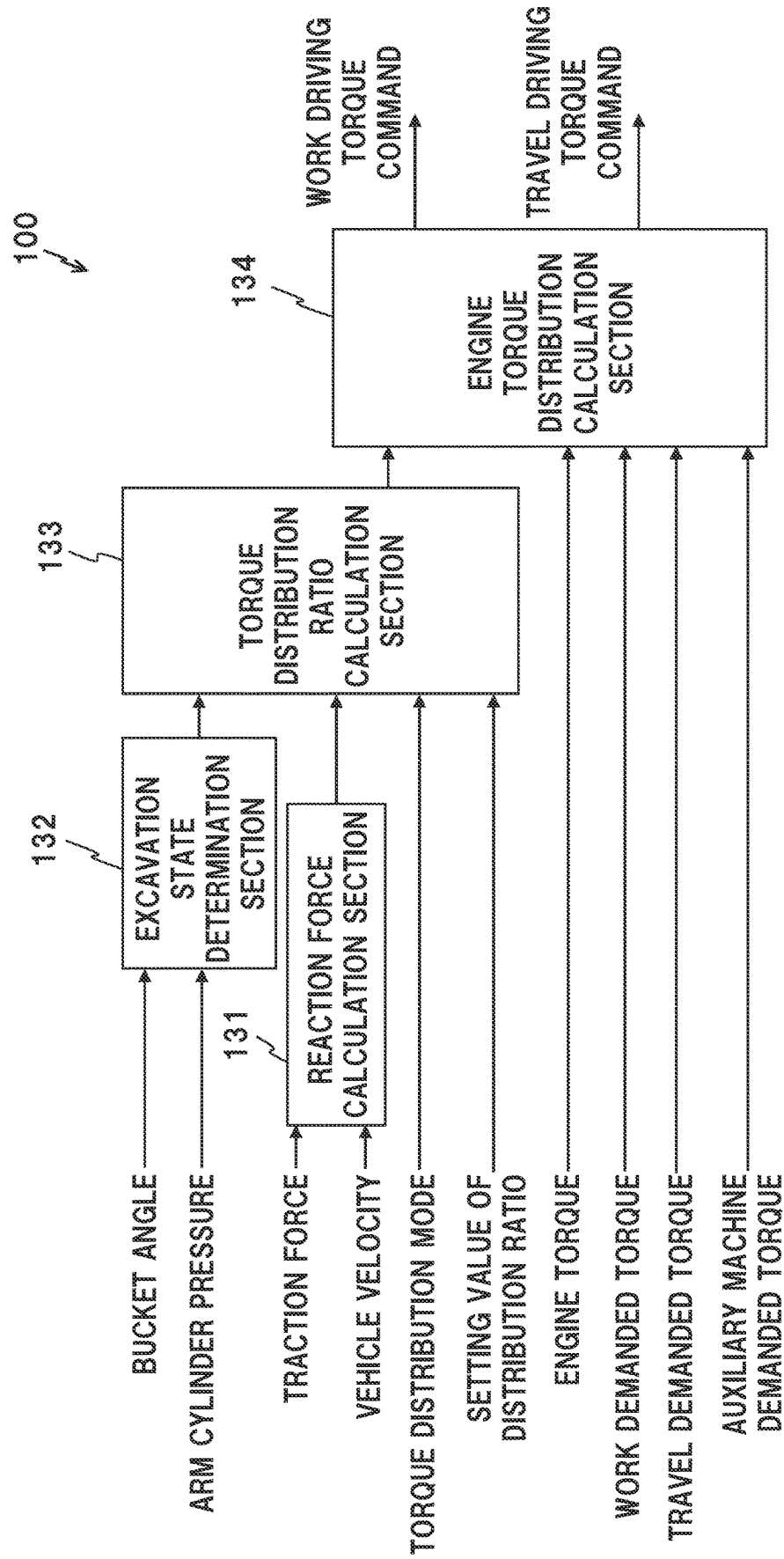


FIG. 5

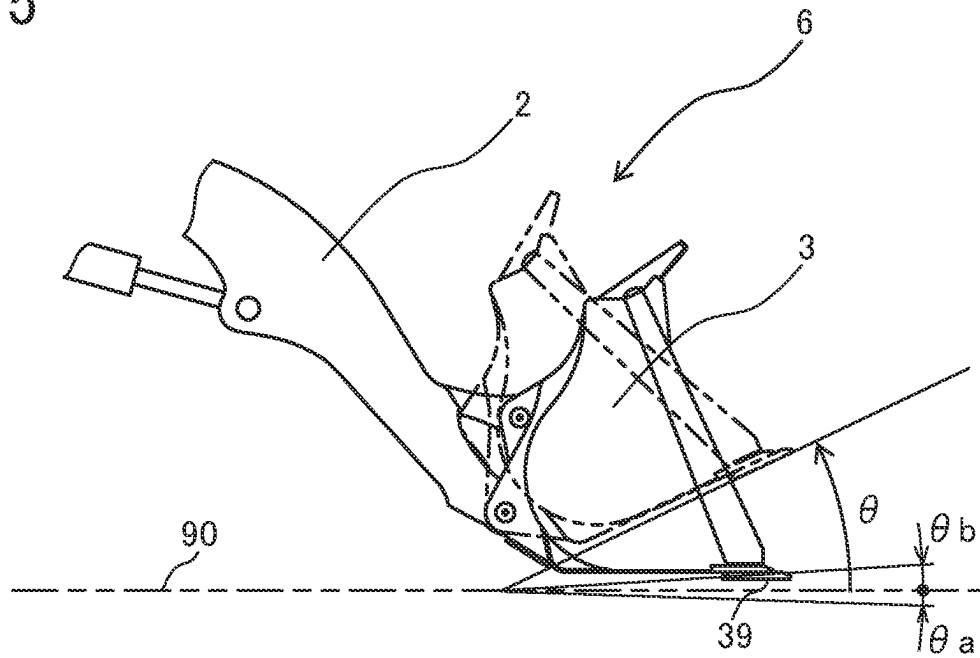


FIG. 6

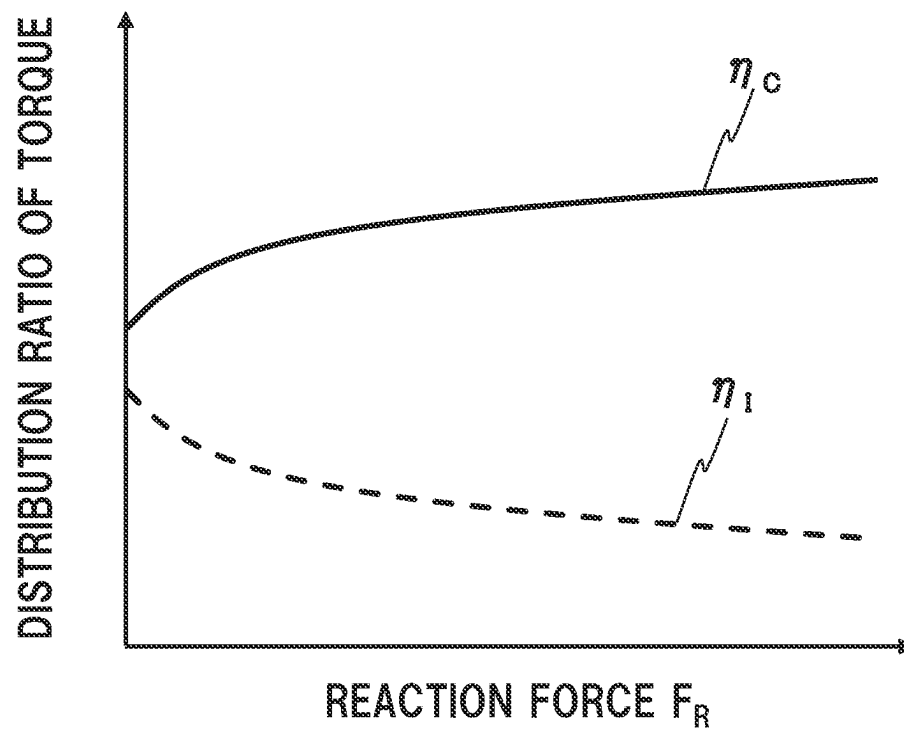


FIG. 7A

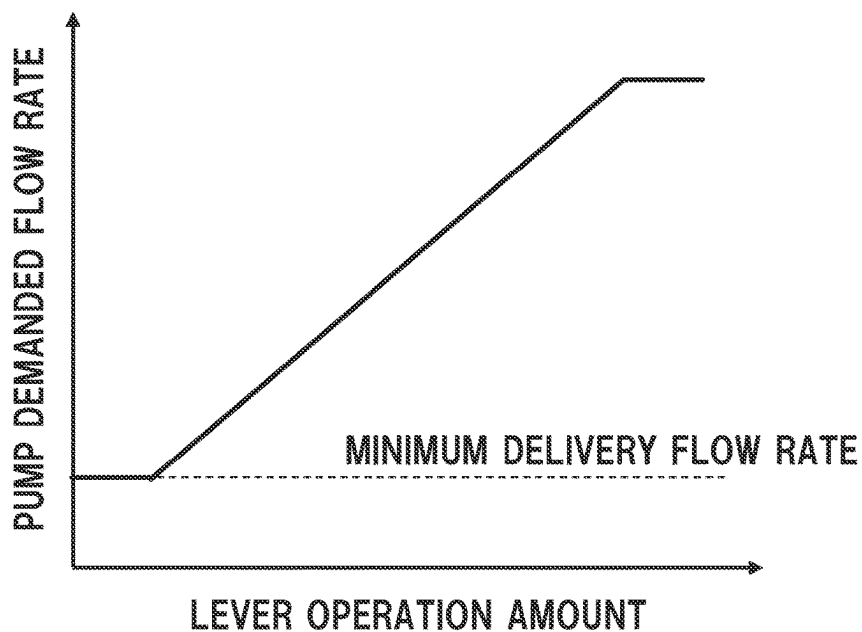


FIG. 7B

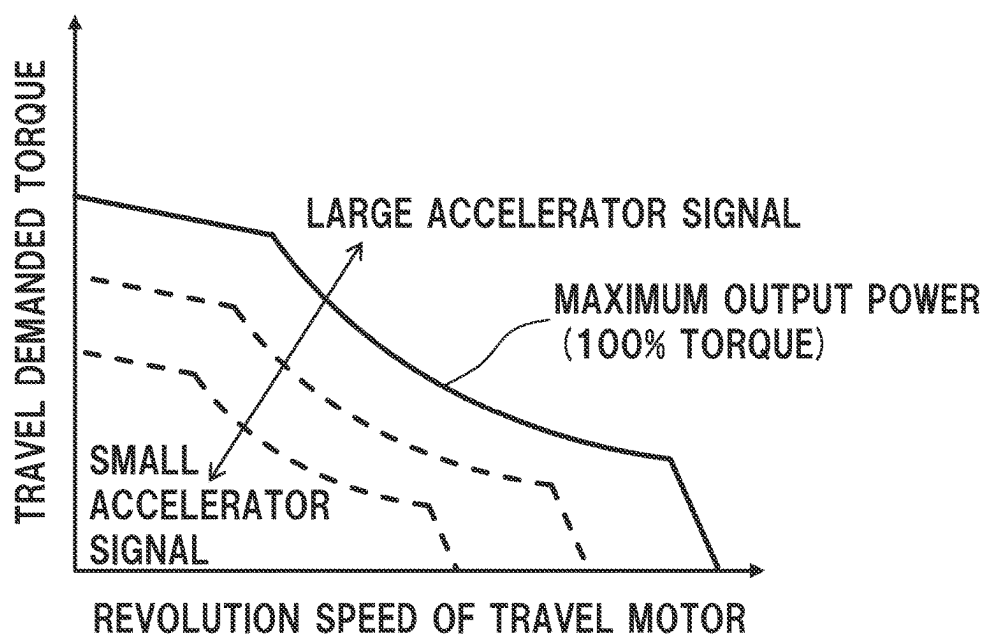


FIG. 8

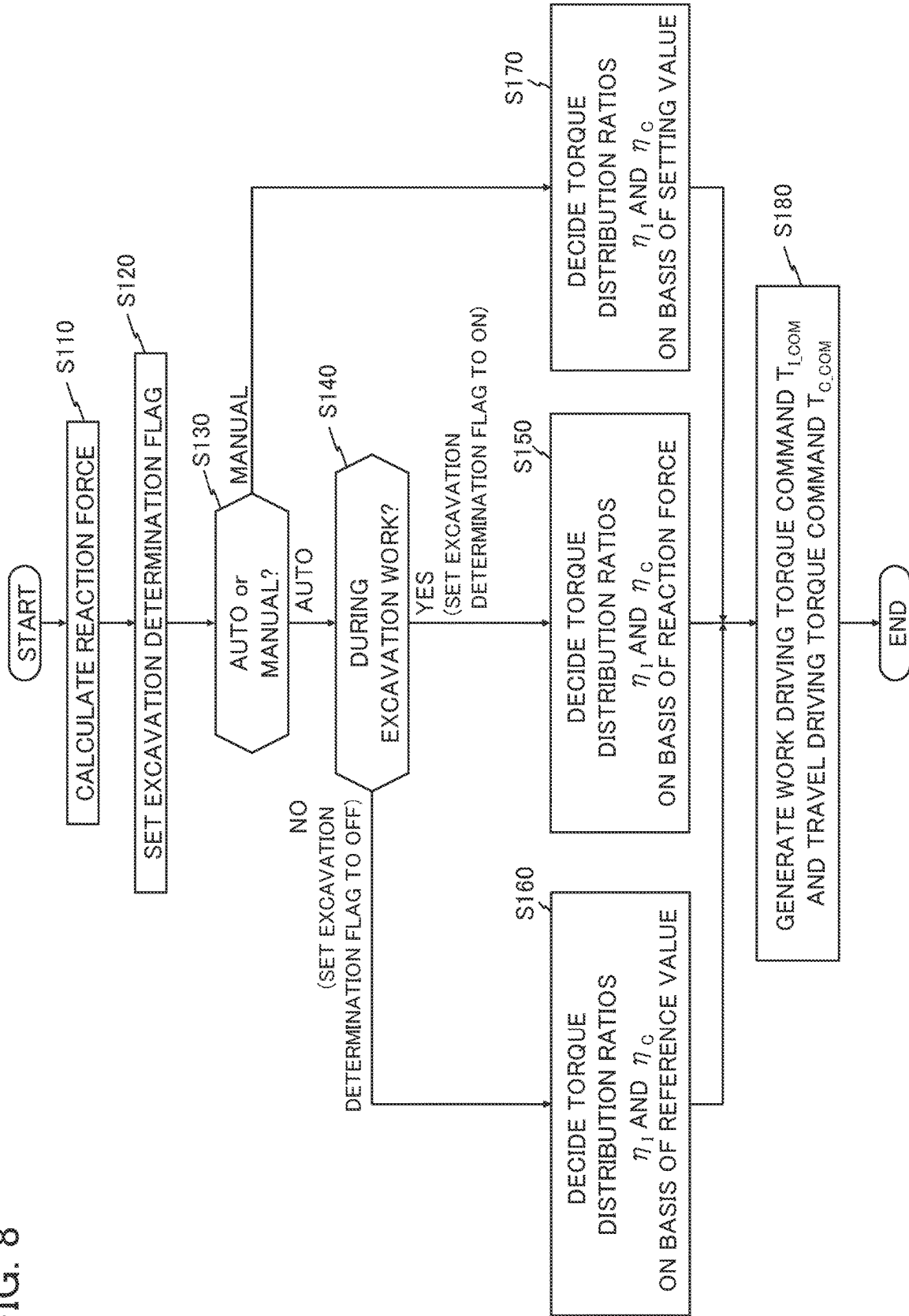


FIG. 9

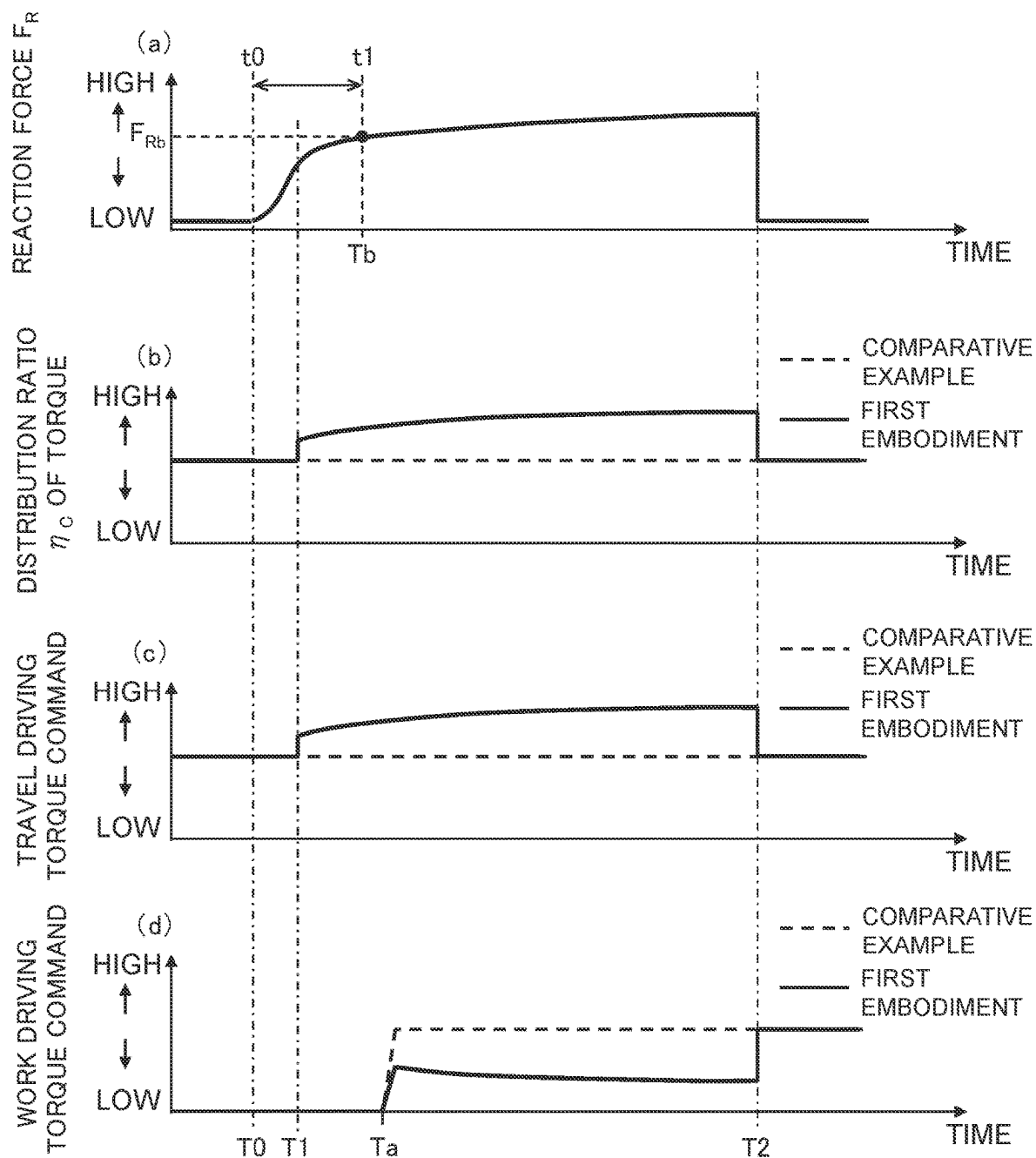


FIG. 10

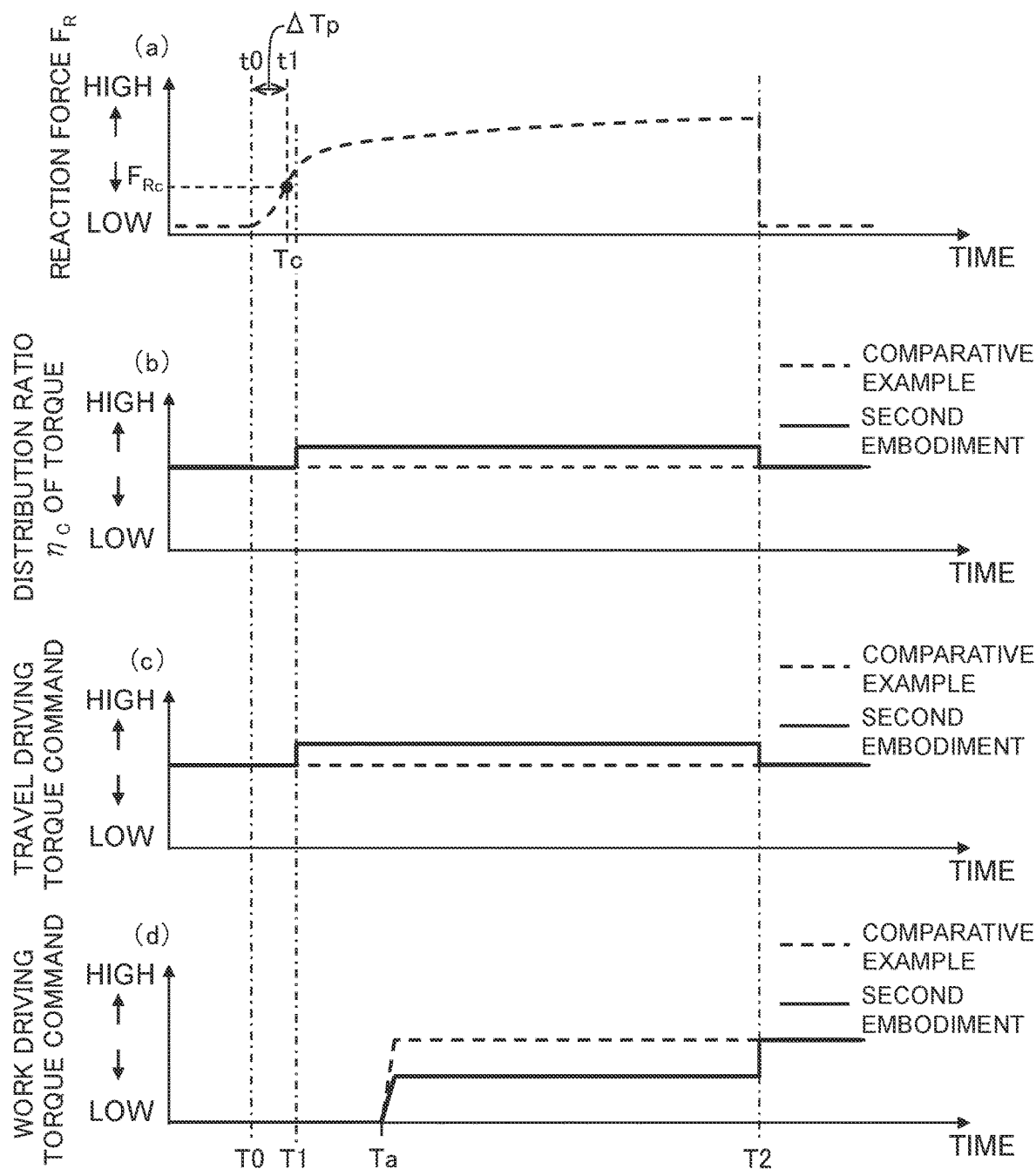


FIG. 11

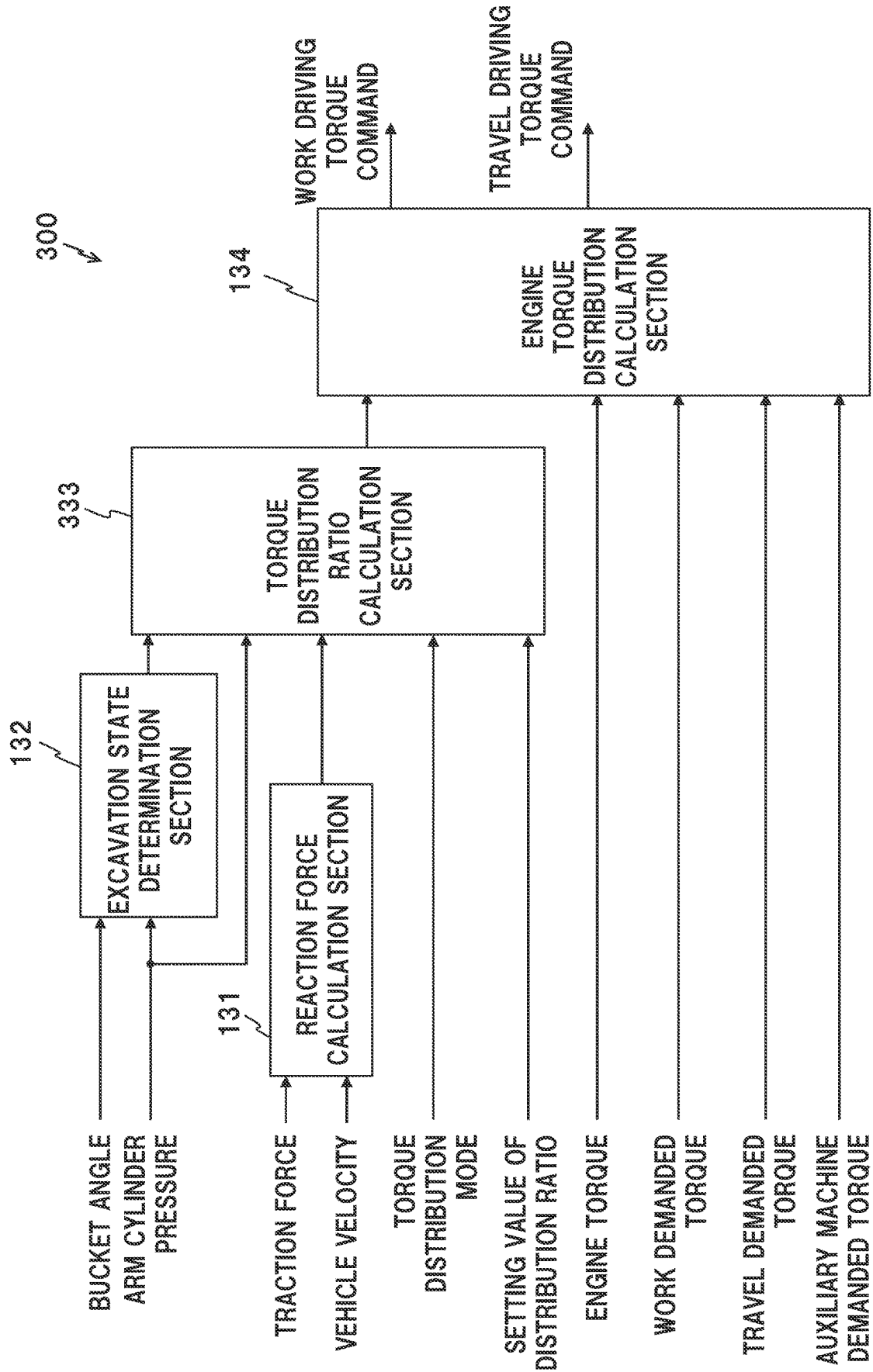


FIG. 12

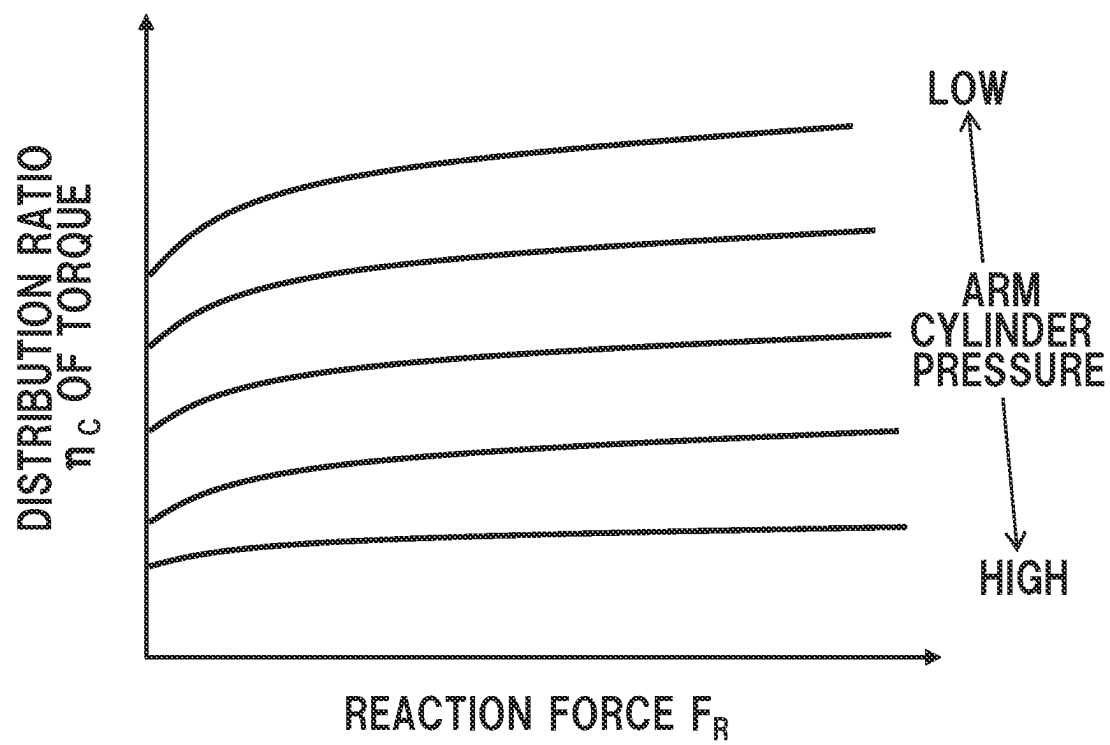


FIG. 13

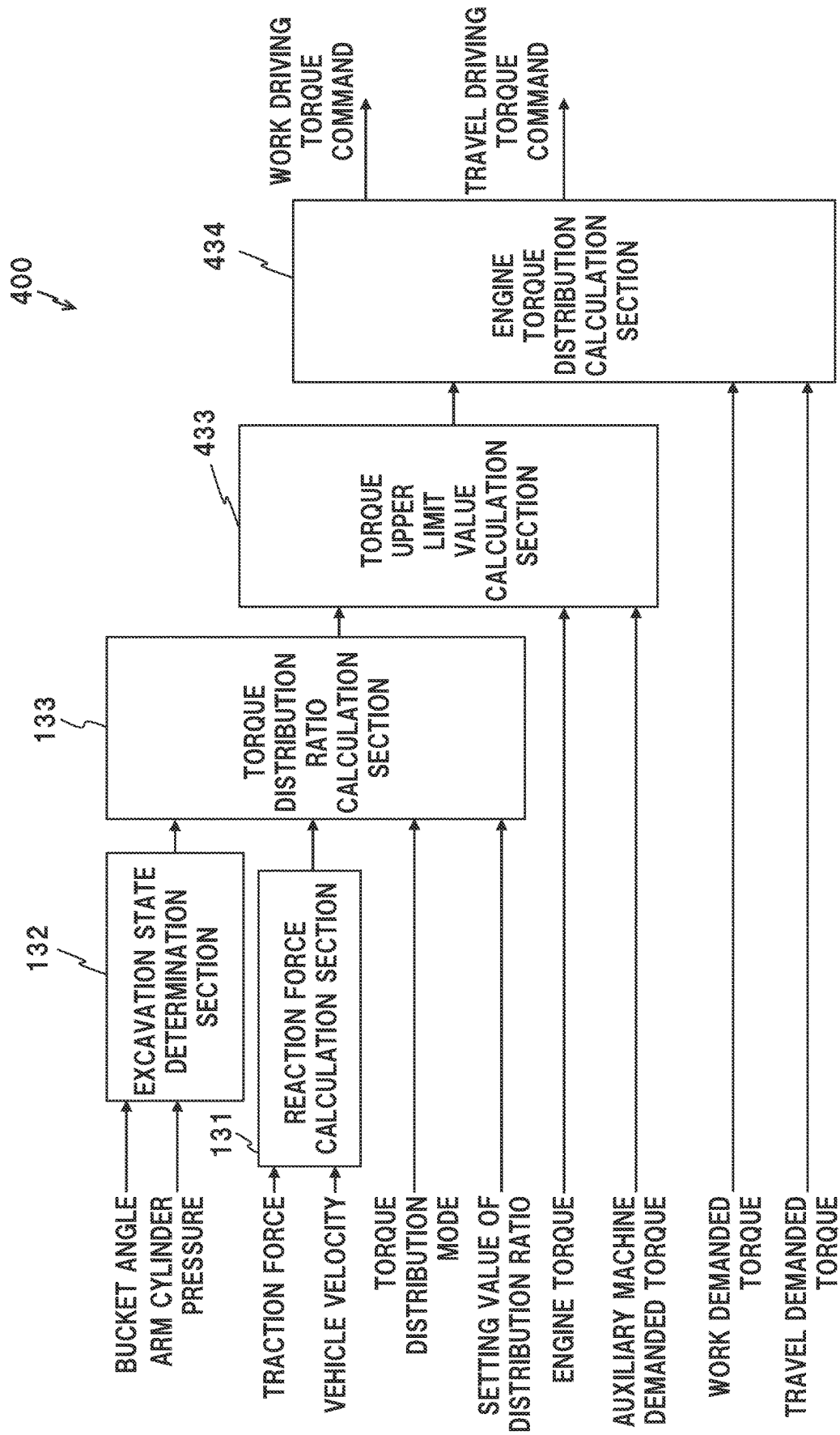
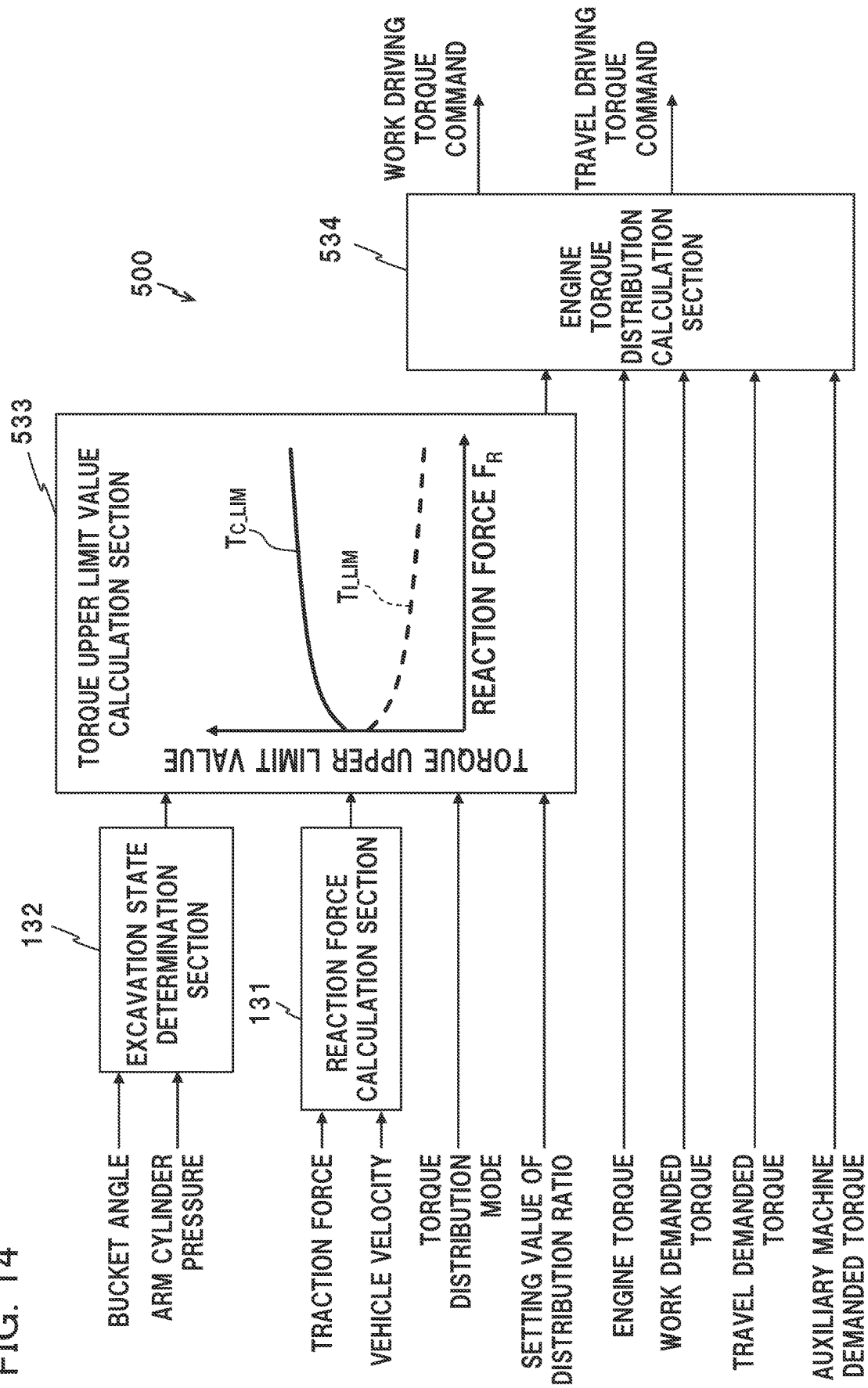


FIG. 14



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**WORK VEHICLE HAVING A CONTROLLER
THAT CONTROLS THE HYDRAULIC
CYLINDERS AND THE ELECTRICALLY
DRIVEN MOTOR ON A BASIS OF A
DISTRIBUTION RATIO OF A FIRST
TORQUE AND A DISTRIBUTION RATIO OF
A SECOND TORQUE**

TECHNICAL FIELD

The present invention relates to a work vehicle.

BACKGROUND ART

A work vehicle including a travel device for moving a vehicle body and a work device having a bucket and arms for excavating sediment and the like has been known. Such a work vehicle performs excavation work by distributing the power of an engine to the travel device and the work device. At this time, if the travel driving force is large and the work driving force is too small, it becomes difficult to lift the bucket, and the work efficiency is reduced. On the other hand, if the travel driving force is small and the work driving force is too large, the work efficiency is reduced because the bucket cannot be sufficiently penetrated into the sediment or the bucket is lifted before the sediment is sufficiently put in the bucket.

Patent Document 1 discloses a method for automatically setting, on the basis of an average bucket weight after excavation work, the distribution ratios (characteristics P, N, and L) of the travel driving force and the work driving force during the subsequent excavation work.

PRIOR ART DOCUMENT

Patent Document

Patent Document 1: JP-2012-233521-A

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

According to the technique described in Patent Document 1, in a case where the excavation work is repeatedly performed, the present excavation work is performed on the basis of the distribution ratios of the travel driving force and the work driving force set on the basis of the average bucket weight after the previous excavation work. Thus, in a case where the hardness of an excavation target targeted for the present excavation work is different from the hardness of an excavation target targeted for the previous excavation work, there is a risk that the set distribution ratios deviate from the ranges of the distribution ratios suitable for the excavation of the present excavation target, and the efficiency of the excavation work is reduced.

An object of the present invention is to improve the efficiency of excavation work.

Means for Solving the Problem

A work vehicle according to an aspect of the present invention includes: an engine that is mounted on a vehicle body; a hydraulic pump that is driven by the engine; hydraulic cylinders that are extended and contracted by pressurized fluid delivered from the hydraulic pump; a work device that is moved according to the extension/contraction

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operations of the hydraulic cylinders; a travel device that is driven independently of the work device; an electrically driven motor that is driven by electric power generated by the engine to operate the travel device; and a controller that controls the hydraulic cylinders and the electrically driven motor. The controller controls output power of the hydraulic pump and output power of the electrically driven motor by changing distribution ratios of a first torque consumed by the work device and a second torque consumed by the travel device among torques output by the engine, on the basis of a reaction force received by the vehicle body.

Advantage of the Invention

According to the present invention, the efficiency of excavation work can be improved.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a side view of a wheel loader.

FIG. 2 is a system configuration diagram of the wheel loader.

FIG. 3 is diagrams for explaining the basic excavation work of the wheel loader.

FIG. 4 is a functional block diagram of a main controller according to a first embodiment.

FIG. 5 is a diagram for explaining a bucket angle θ .

FIG. 6 is a diagram for showing a torque distribution ratio table according to the first embodiment.

FIG. 7A is a diagram for showing an example of a pump demanded map used to calculate a pump demanded flow rate.

FIG. 7B is a diagram for showing an example of a torque map of a travel motor used to calculate a travel demanded torque.

FIG. 8 is a flowchart for showing the content of a torque decision process executed by the main controller.

FIG. 9 is time charts for showing the operation of the main controller according to the first embodiment.

FIG. 10 is time charts for showing the operation of a main controller according to a second embodiment.

FIG. 11 is a functional block diagram of a main controller according to a third embodiment.

FIG. 12 is a diagram for showing a torque distribution ratio table according to the third embodiment.

FIG. 13 is a functional block diagram of a main controller according to a fourth embodiment.

FIG. 14 is a functional block diagram of a main controller according to a fifth embodiment.

**MODES FOR CARRYING OUT THE
INVENTION**

Hereinafter, embodiments of the present invention will be described with reference to the drawings. It should be noted that an example in which a work vehicle is an electrically-driven wheel loader will be described in the embodiments.

First Embodiment

A work vehicle according to a first embodiment of the present invention will be described with reference to FIG. 1 to FIG. 9. FIG. 1 is a side view of a wheel loader. As shown in FIG. 1, a wheel loader 1 includes a vehicle body 8 mounted with a travel device 11 and an articulated work device 6 attached to the front of the vehicle body 8. The vehicle body 8 is an articulated steering type (vehicle body

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bending type) vehicle body and has a front vehicle body 8A, a rear vehicle body 8B, and a center joint 10 coupling the front vehicle body 8A and the rear vehicle body 8B to each other.

On the rear vehicle body 8B, an operation room 12 and an engine room 16 are mounted in the front and the rear, respectively. An engine 20 (see FIG. 2), hydraulic pumps 30A, 30B, and 30C (see FIG. 2) driven by the engine 20, and hydraulic equipment such as valves are mounted in the engine room 16.

FIG. 2 is a system configuration diagram of the wheel loader 1. As shown in FIG. 2, the wheel loader 1 includes the engine 20, a power generation motor 40 mechanically connected to the engine 20, the hydraulic pumps 30A, 30B, and 30C mechanically connected to the engine 20, the work device 6 driven by a hydraulic operating fluid delivered from the hydraulic pump 30A, a front device control section 31 controlling the operation of the work device 6, a brake device 21 driven by a hydraulic operating fluid delivered from the hydraulic pump 30B, a brake control section 32 controlling the operation of the brake device 21, a steering device 22 driven by a hydraulic operating fluid delivered from the hydraulic pump 30C, a steering control section 33 controlling the steering device 22, and the travel device 11 driven by the electric power generated by the power generation motor 40.

The work device 6 and the travel device 11 are driven independently of each other by the power of the engine 20. The engine 20 is configured using, for example, an internal combustion engine such as a diesel engine.

As shown in FIG. 1, the work device 6 is attached to the front vehicle body 8A. The work device 6 has lift arms (hereafter, simply referred to as arms) 2 rotatably attached to the front vehicle body 8A and a bucket 3 rotatably attached to the arms 2. The arms 2 are moved according to the extension/contraction operation of arm cylinders 4, which are hydraulic cylinders, and the bucket 3 is moved according to the extension/contraction operation of a bucket cylinder 5, which is a hydraulic cylinder. It should be noted that one arm 2 and one arm cylinder 4 are provided on each of the left and right of the front vehicle body 8A. In addition, in the embodiment, a Z-link type (bell crank type) link mechanism is employed as a link mechanism for operating the bucket 3.

The hydraulic cylinders 4 and 5 are extended and contracted by the hydraulic operating fluid (pressurized fluid) delivered from the hydraulic pump 30A that is rotated by the torque output by the engine 20 (see FIG. 2).

The travel device 11 has front wheels 7A (tires 7) attached to the front vehicle body 8A, rear wheels 7B (tires 7) attached to the rear vehicle body 8B, and a power transmission device that transmits power from the travel motor 43 to the tires 7. The power transmission device includes an axle, a differential device, a propeller shaft, and the like.

The travel motor 43 is an electrically driven motor that is rotationally driven by the electric power generated by the power generation motor 40 rotated by the torque output from the engine 20 and operates the travel device 11. It should be noted that the wheel loader 1 is turned by the steering device 22 having a pair of left and right hydraulic cylinders (hereafter, referred to as steering cylinders) 15 provided so as to couple the front vehicle body 8A and the rear vehicle body 8B to each other.

As shown in FIG. 2, the hydraulic pumps 30A, 30B, and 30C are mechanically connected to the engine 20 and the power generation motor 40. The hydraulic pumps 30A, 30B, and 30C are driven by the torque output from the engine 20 to deliver the hydraulic operating fluids as working fluids. It

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should be noted that in a case where the power generation motor 40 functions as an electric motor, the hydraulic pumps 30A, 30B, and 30C are driven by the torque output from the engine 20 and the power generation motor 40.

The pressure, flow rate, and flow direction of the hydraulic operating fluid delivered from the hydraulic pump 30A are controlled by the front device control section 31. The pressure, flow rate, and flow direction of the hydraulic operating fluid delivered from the hydraulic pump 30B are controlled by the brake control section 32. The pressure, flow rate, and flow direction of the hydraulic operating fluid delivered from the hydraulic pump 30C are controlled by the steering control section 33.

The wheel loader 1 includes a main controller 100 that is a controller for controlling the hydraulic cylinders 4 and 5 for operating the work device 6 and the travel motor 43 for operating the travel device 11, a power generation inverter (inverter for the power generation motor) 41 that controls the power generation motor 40 on the basis of a power generation voltage command input from the main controller 100, a travel inverter (inverter for the travel motor) 42 that controls the torque of the travel motor 43 on the basis of a travel driving torque command input from the main controller 100, and various operation members (50 to 59) provided in the operation room 12.

In the operation room 12, provided are an arm operation lever 52 for driving the arms 2, a bucket operation lever 53 for driving the bucket 3, a forward/backward switch 51 that is a forward/backward switching device for switching the forward (F) and backward (R) of the vehicle body 8, an accelerator pedal 58 for accelerating the vehicle body 8, a brake pedal 59 for decelerating the vehicle body 8, a steering wheel 55 for instruction on the left/right advancing direction of the vehicle body 8, a mode switching switch 57 that is a switching device for switching a torque distribution mode to either a MANUAL mode or an AUTO mode, a torque distribution ratio setting dial 54 for manually setting the ratio at which the output torque of the engine 20 is distributed to the work device 6 and the travel device 11, and a parking brake switch 56 for operating the parking brake.

The main controller 100 is configured using a microcomputer including a CPU (Central Processing Unit) 104 as an operating circuit, a ROM (Read Only Memory) 105 and a RAM (Random Access Memory) 106 as storage devices, an input interface 107, an output interface 108, and other peripheral circuits. It should be noted that the main controller 100 may be configured using one microcomputer or a plurality of microcomputers.

The ROM 105 of the main controller 100 is a nonvolatile memory such as an EEPROM and stores programs that can execute various computations. That is, the ROM 105 of the main controller 100 is a storage medium that can read programs for realizing the functions of the embodiment. The RAM 106 is a volatile memory and a work memory that directly inputs and outputs data with the CPU 104. The RAM 106 temporarily stores necessary data while the CPU 104 computes and executes a program. It should be noted that the main controller 100 may further include a storage device such as a flash memory or a hard disk drive.

The CPU 104 is a processing device that expands the program stored in the ROM 105 into the RAM 106 to execute computations, and performs a predetermined computation process for signals taken from the input interface 107, the ROM 105, and the RAM 106 according to the program.

An operation signal and a sensor signal are input to the input interface 107. The input interface 107 converts the

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input signal such that the CPU 104 can compute. The output interface 108 generates a signal for output according to the computation result in the CPU 104, and outputs the signal to the front device control section 31, the brake control section 32, the steering control section 33, the power generation inverter 41, the travel inverter 42, and the like.

The main controller 100 comprehensively controls the front device control section 31, the brake control section 32, the steering control section 33, the power generation inverter 41, and the travel inverter 42 on the basis of the operation signals input by an operator's operation and the sensor signals sensed by various sensors.

The operation signals input to the main controller 100 include an accelerator signal output from the accelerator pedal 58 and indicating the operation amount of the accelerator pedal 58, a brake signal output from the brake pedal 59 and indicating the operation amount of the brake pedal 59, an arm signal output from the arm operation lever 52 and indicating the operation amount of the arm operation lever 52, a bucket signal output from the bucket operation lever 53 and indicating the operation amount of the bucket operation lever 53, a steering signal output from the steering wheel 55 and indicating the operation amount of the steering wheel 55, an advancing direction signal output from the forward/backward switch 51 and indicating the operation position of the forward/backward switch 51, and the like. In addition, the operation signals input to the main controller 100 include a mode switching signal output from the mode switching switch 57 and indicating the operation position of the mode switching switch 57, and a ratio setting signal output from the torque distribution ratio setting dial 54 and indicating the operation position of the torque distribution ratio setting dial 54.

The sensor signals input to the main controller 100 include a signal indicating the angle sensed by an arm relative angle sensor 62 provided on a coupling shaft coupling the vehicle body 8 and the arms 2 to each other, and a signal indicating the angle sensed by a bucket relative angle sensor 63 provided on a coupling shaft coupling the arms 2 and the bucket 3 to each other. The arm relative angle sensor 62 is a potentiometer that senses the relative angle (inclined angle) of the arms 2 to the vehicle body 8 and outputs a signal indicating the sensed angle to the main controller 100. The bucket relative angle sensor 63 is a potentiometer that senses the relative angle (inclined angle) of the bucket 3 to the arms 2 and outputs a signal indicating the sensed angle to the main controller 100. Since the angle of the vehicle body 8 relative to the ground (travel surface) is constant, the angle sensed by the arm relative angle sensor 62 can be said to correspond to the relative angle (inclined angle) of the arms 2 to the ground.

In addition, the sensor signals input to the main controller 100 include a signal indicating the vehicle velocity sensed by a vehicle velocity sensor 61. The vehicle velocity sensor 61 senses the vehicle velocity (travel velocity) of the wheel loader 1 and outputs a signal indicating the sensed vehicle velocity to the main controller 100. Further, the sensor signals input to the main controller 100 include signals indicating the revolution speeds of the engine 20, the power generation motor 40, the hydraulic pumps 30A, 30B, and 30C, and the travel motor 43 sensed by a plurality of revolution speed sensors, and signals indicating the delivery pressures of the hydraulic pumps 30A, 30B, and 30C, the pressure (load pressure) of the hydraulic cylinder, and the like sensed by a plurality of pressure sensors.

The main controller 100 outputs a front device control command on the basis of the operation directions and the

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operation amounts of the operation levers 52 and 53. The front device control section 31, on the basis of the front device control command from the main controller 100, adjusts the pressure, velocity, and direction of the hydraulic operating fluid delivered from the hydraulic pump 30A to operate the arm cylinders 4 and the bucket cylinder 5. The front device control section 31 has a direction control valve for controlling the flow of the hydraulic operating fluid delivered from the hydraulic pump 30A, a solenoid valve for generating a pilot pressure input into the pilot chamber of the direction control valve, and the like.

The main controller 100 outputs a brake control command on the basis of the operation amount of the brake pedal 59 and the operation position of the parking brake switch 56. The brake control section 32, on the basis of the brake control command from the main controller 100, adjusts the pressure, velocity, and direction of the hydraulic operating fluid delivered from the hydraulic pump 30B to operate hydraulic cylinders 17 and 18 for functioning the brake 13 and the parking brake 14. The brake control section 32 has a direction control valve for controlling the flow of the hydraulic operating fluid delivered from the hydraulic pump 30B, a solenoid valve for generating a pilot pressure input into the pilot chamber of the direction control valve, and the like.

The main controller 100 outputs a steering control command on the basis of the operation direction and the operation amount of the steering wheel 55. The steering control section 33, on the basis of the steering control command from the main controller 100, adjusts the pressure, velocity, and direction of the hydraulic operating fluid delivered from the hydraulic pump 30C to operate the steering cylinders 15. The steering control section 33 has a direction control valve for controlling the flow of the hydraulic operating fluid delivered from the hydraulic pump 30C, a solenoid valve for generating a pilot pressure input into the pilot chamber of the direction control valve, and the like.

As described above, in the embodiment, the hydraulic pumps 30A, 30B, and 30C are driven by the torque output from the engine 20, and the work device 6, the brake device 21, and the steering device 22 are driven by the hydraulic operating fluids delivered from the hydraulic pumps 30A, 30B, and 30C.

The power generation inverter 41 and the travel inverter 42 are connected to each other via a DC section (DC bus) 44. It should be noted that the wheel loader 1 according to the embodiment does not include a power storage device connected to the DC section 44. The power generation inverter 41, on the basis of a power generation voltage command from the main controller 100, controls the bus voltage of the DC section 44 by using the electric power supplied from the power generation motor 40. The travel inverter 42, on the basis of the travel driving torque command of the main controller 100, drives the travel motor 43 by using the electric power of the DC section 44.

As described above, in the embodiment, the power generation motor 40 is driven by the torque output from the engine 20, and the travel motor 43 is driven by the electric power generated by the power generation motor 40.

When the arm operation lever 52 is operated, the arms 2 are rotated (moved up and down) in the up and down directions by the extension/contraction operation of the arm cylinders 4. When the bucket operation lever 53 is operated, the bucket 3 is rotated (crowding operation or dump operation) in the up and down directions by the extension/contraction operation of the bucket cylinder 5.

When the steering wheel **55** is operated, the front vehicle body **8A** is refracted (turned) left and right with the center joint **10** in the center against the rear vehicle body **8B**, according to the extension/contraction operation of the steering cylinders **15**. The tires **7** are rotated by the driving of the travel motor **43** to move the wheel loader **1** forward and backward.

When the accelerator pedal **58** is stepped on in a state where the forward/backward switch **51** is operated to the forward side, the tires **7** rotate in the forward direction and the vehicle body **8** travels forward. When the accelerator pedal **58** is stepped on in a state where the forward/backward switch **51** is operated to the backward side, the tires **7** rotate in the backward direction and the vehicle body **8** travels backward.

The mode switching switch **57** is a mode switch operation section that can be manually switched between an AUTO mode (first control mode) in which the work driving torque and the travel driving torque are controlled on the basis of a reaction force F_R acting on the vehicle body **8** of the wheel loader **1** and a MANUAL mode (second control mode) in which the work driving torque and the travel driving torque are controlled such that the ratios of the work driving torque and the travel driving torque are held at predetermined ratios regardless of the reaction force F_R .

It should be noted that the work driving torque corresponds to the torque (first torque) consumed by the work device **6** among those output from the engine **20**. That is, in the embodiment, the work driving torque corresponds to the input torque of the hydraulic pump **30A**. The travel driving torque corresponds to the torque (second torque) consumed by the travel device **11** among those output from the engine **20**. That is, in the embodiment, the travel driving torque corresponds to the output torque of the travel motor **43**.

The mode switching switch **57** has an AUTO mode position and a MANUAL mode position. Where the mode switching switch **57** is operated to the AUTO mode position, a signal indicating that the AUTO mode is selected is output to the main controller **100**. In this case, the main controller **100** sets the AUTO mode as the torque distribution mode. Where the mode switching switch **57** is operated to the MANUAL mode position, a signal indicating that the MANUAL mode is selected is output to the main controller **100**. In this case, the main controller **100** sets the MANUAL mode as the torque distribution mode.

The torque distribution ratio setting dial **54** is an operation member for setting a setting value η_{CS} of the torque distribution ratio used when the torque distribution mode is set to the MANUAL mode. The main controller **100**, on the basis of the operation position of the torque distribution ratio setting dial **54**, sets the setting value η_{CS} of the distribution ratio of the travel driving torque. Further, the main controller **100** sets a setting value η_{TS} of the distribution ratio of the work driving torque. The main controller **100** subtracts the setting value $\eta_{CS}[\%]$ of the distribution ratio of the travel driving torque from 100[%] to calculate the setting value η_{TS} of the distribution ratio of the work driving torque ($100 - \eta_{CS} = \eta_{TS} [\%]$). The main controller **100** stores the setting values η_{CS} and η_{TS} of the torque distribution ratios in the ROM **105**.

Next, the basic excavation work of the wheel loader **1** will be described with reference to FIG. **3**. In the excavation work of the wheel loader **1**, the wheel loader **1** is first advanced toward an excavation target **91** such as a sediment pile or a natural ground as shown in FIG. **3(a)**. Next, as shown in FIG. **3(b)**, the bucket **3** is penetrated into the excavation target **91** in such a manner as to plunge into the

excavation target **91**, and the arms **2** and the bucket **3** are operated to put objects to be carried such as sediment into the bucket **3**. Finally, as shown in FIG. **3(c)**, the excavation work is completed by scooping up (performing the crowding operation) the bucket **3** toward the operator so as not to spill the objects to be carried such as sediment put in the bucket **3**. After the excavation work is completed, the wheel loader **1** is moved backward once and advanced toward a conveying vehicle such as a dump truck. At this time, the operator operates the steering wheel **55** to advance the wheel loader **1** toward the dump truck while raising the arms **2** by operating the arm operation lever **52**. After stopping the wheel loader **1** in front of the dump truck, the operator loads the objects to be carried in the bucket **3** onto the cargo bed of the dump truck (that is, releases the sediment in the bucket **3**) by operating the bucket operation lever **53** to allow the bucket **3** to perform the dump operation. When the loading work is completed, the operator moves the wheel loader **1** backward again and returns to the original position.

A series of work including the excavation work and the loading work accounts for the majority of the whole work time of the wheel loader **1**. Therefore, in order to improve the work efficiency of the wheel loader **1**, it is effective to improve the efficiency of the series of work. It should be noted that the work efficiency corresponds to, for example, the weight [ton/h] of the excavated material loaded onto the conveying vehicle per unit time in the series of work including the excavation work and the loading work, and a larger weight means that a more amount of materials can be excavated in a shorter time.

Here, if the balance between the driving force (travel driving force) of the travel device **11** and the driving force (work driving force) of the work device **6** is poor, the efficiency of the excavation work is reduced. For example, where the travel driving force is insufficient and the work driving force is excessive, the work efficiency is reduced because the bucket **3** cannot be sufficiently penetrated into the excavation target **91** or the bucket **3** is lifted above the excavation target **91** before the sediment is sufficiently put in the bucket **3**. On the other hand, where the work driving force is insufficient and the travel driving force is excessive, the work efficiency is reduced because it takes time to lift the bucket **3**. As described above, if the balance between the travel driving force and the work driving force is poor, the work efficiency is reduced.

Accordingly, in the embodiment, the efficiency of the excavation work in the series of work is improved by appropriately balancing the driving force (travel driving force) of the travel device **11** required to appropriately penetrate the bucket **3** into the excavation target **91** and the driving force (work driving force) of the work device **6** required to load a large amount of sediment into the bucket **3** in a short time.

It should be noted that the appropriate balance between the travel driving force and the work driving force changes depending on the hardness of the excavation target **91**. As the hardness of the excavation target **91** is greater, a larger travel driving force is required. The influence of the hardness of the excavation target **91** appears in the reaction force received by the vehicle body **8** of the wheel loader **1** from the excavation target **91** during the excavation work. The reaction force acting on the vehicle body **8** of the wheel loader **1** becomes larger as the hardness of the excavation target **91** is greater. In the embodiment, the work efficiency is improved by adjusting the balance between the travel driving force and the work driving force on the basis of the

reaction force acting on the vehicle body **8** of the wheel loader **1** during the excavation work.

FIG. 4 is a functional block diagram of the main controller **100**. As shown in FIG. 4, the main controller **100**, by executing the programs stored in the ROM **105**, functions as a reaction force calculation section **131** that calculates the reaction force F_R received by the vehicle body **8** of the wheel loader **1**, an excavation state determination section **132** that determines whether or not the wheel loader **1** is in the state of performing the excavation work, a torque distribution ratio calculation section **133** that decides the distribution ratios η_I and η_C of the work driving torque and the travel driving torque on the basis of the reaction force F_R calculated by the reaction force calculation section **131**, and an engine torque distribution calculation section **134** that decides a work driving torque command for controlling the work driving torque and a travel driving torque command for controlling the travel driving torque on the basis of the distribution ratios η_I and η_C decided by the torque distribution ratio calculation section **133**.

The reaction force calculation section **131**, on the basis of the traction force F_P and the vehicle velocity v of the wheel loader **1**, calculates the reaction force F_R received by the vehicle body **8** of the wheel loader **1**. An example of a method for calculating the reaction force F_R will be described. The reaction force F_R is obtained by the following equation (1) using a relation between momentum and an impulse.

[Equation 1]

$$F_R = F_P - \frac{mv_1 - mv_0}{t_1 - t_0} \quad (1)$$

Here, m is the mass of the wheel loader **1**, t_0 is a reference time, t_1 is a time after a predetermined time elapses from the reference time t_0 , v_0 is the vehicle velocity of the wheel loader **1** at the reference time t_0 , and v_1 is the vehicle velocity of the wheel loader **1** at the time t_1 . It should be noted that although the vehicle velocity of the wheel loader **1** is sensed by the vehicle velocity sensor **61** in the embodiment, a rotary encoder that senses the revolution speed of a shaft configuring the power transmission device may be provided to compute the vehicle velocity on the basis of information sensed by the rotary encoder.

The traction force F_P correlated with the travel driving force generated by the travel motor **43** and is obtained by the following equation (2) on the basis of, for example, the output torque (travel driving torque) T_m of the travel motor **43** at the time t_1 , a general reduction ratio λ , and the diameter D of the tire **7**.

[Equation 2]

$$F_P = \frac{T_m \cdot \lambda}{Dt} \cdot c \quad (2)$$

It should be noted that the general reduction ratio A is the ratio of the revolution speed of the engine **20** to the revolution speed of the tires **7**, and where a transmission is provided, it is calculated by multiplying the gear ratio of the transmission by the reduction ratio (differential ratio). In the equation, c is a factor for unit conversion. The travel driving torque (motor output torque) T_m may be sensed by a torque sensor or computed from the motor current sensed by a current sensor.

The reaction force F_R calculated during the excavation work is dominated by the repulsive force acting on the wheel loader **1** from the excavation target **91** and is affected by the hardness of the excavation target **91**. The reference time t_0 is the time when the wheel loader **1** plunged into the excavation target **91** (the time when the bucket **3** started penetrating the excavation target **91**).

The main controller **100** determines whether or not the wheel loader **1** has plunged into the excavation target **91** (that is, whether or not the bucket **3** has started penetrating the excavation target **91**) on the basis of the operation amount of the accelerator pedal **58** and the time change ratio of the vehicle velocity v of the wheel loader **1**. When the main controller **100** determines that the wheel loader **1** has plunged into the excavation target **91**, a plunge flag is set to ON. In addition, the main controller **100** sets the time when the plunge flag has been set to ON as the reference time t_0 . It should be noted that the plunge flag is set to OFF when an excavation determination flag to be described later is set to OFF.

The main controller **100** calculates the time change ratio D ($= (v_b - v_a) / (t_b - t_a)$) of the vehicle velocity v by dividing a difference ($v_b - v_a$) between the previous value v_a and the current value v_b of the vehicle velocity v , which is repeatedly sensed in a predetermined control cycle, by a time Δt ($= t_b - t_a$) from a time t_a when the previous value v_a has been sensed to a time t_b when the current value v_b has been sensed. When the operation amount of the accelerator pedal **58** is equal to or more than a predetermined threshold value and the time change ratio D of the vehicle velocity v is equal to or more than a predetermined threshold value, the main controller **100** determines that the wheel loader **1** has plunged into the excavation target **91**. When the operation amount of the accelerator pedal **58** is less than the predetermined threshold value or when the time change ratio D of the vehicle velocity v is less than the predetermined threshold value, the main controller **100** determines that the wheel loader **1** has not plunged into the excavation target **91**.

The excavation state determination section **132** determines, on the basis of the angle (hereinafter, also referred to as a bucket angle) θ of the bucket **3** and the pressure (hereinafter, also referred to as a bottom pressure) P_a of the bottom-side hydraulic fluid chamber of the arm cylinder **4**, whether or not the wheel loader **1** is in the state of performing the excavation work. The bottom pressure P_a of the arm cylinder **4** is sensed by a pressure sensor **71** provided in a hydraulic fluid line connecting the bottom-side hydraulic fluid chamber (not shown) of the arm cylinder **4** and the front device control section **31** to each other as shown in FIG. 2. The pressure sensor **71** outputs a sensed signal to the main controller **100**.

FIG. 5 is a diagram for explaining the bucket angle θ . As shown in FIG. 5, the bucket angle θ is the inclined angle of the bucket **3** from a reference surface **90**. In the embodiment, the reference surface **90** is a surface set parallel to the ground (travel surface). In a state where the bottom surface of a blade section **39** of the bucket **3** is parallel to the reference surface **90**, the bucket angle θ is 0° . When the bucket **3** is rotated by the crowding operation, the bucket angle θ increases with the rotation. In other words, when the bucket **3** is rotated by the dump operation, the bucket angle θ decreases with the rotation. The bucket angle θ is calculated by the main controller **100** on the basis of the relative angle of the arms **2** to the reference surface **90** sensed by the arm relative angle sensor **62** and the relative angle of the bucket **3** to the arms **2** sensed by the bucket relative angle sensor **63**.

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The excavation state determination section 132, when the bucket angle θ is within a preset angle range (a lower threshold value θ_a to an upper threshold value θ_b) and the bottom pressure P_a of the arm cylinder 4 is equal to or more than a preset pressure threshold value P_{a0} , determines that the excavation work has been started and sets the excavation determination flag to ON. The excavation determination flag is set to ON when the wheel loader 1 is in the state of performing the excavation work and is set to OFF when the excavation work is finished.

The lower threshold value θ_a and the upper threshold value θ_b defining the angle range are set on the basis of the plunging posture of the work device 6. The plunging posture of the work device 6 is a posture in which the bucket 3 is approximately parallel to the travel surface (ground) in the vicinity of the travel surface (ground). The lower threshold value θ_a and the upper threshold value θ_b are stored in advance in the ROM 105.

The bottom pressure P_a of the arm cylinder 4 rises due to the plunging of the wheel loader 1 into the excavation target 91. The pressure threshold value P_{a0} is stored in advance in the ROM 105. As the pressure threshold value P_{a0} , it is possible to employ, for example, a value about twice the bottom pressure of the arm cylinder 4 supporting the work device 6 whose bucket 3 is empty and which is in the plunging posture.

That is, the excavation state determination section 132 determines that the excavation work has been started when the plunging of the wheel loader 1 into the excavation target 91 is sensed by the rise of the arm cylinder pressure in a state where the work device 6 is in the plunging posture.

The excavation state determination section 132, after it is determined that the excavation work has been started, determines that the excavation work has been finished and sets the excavation determination flag to OFF when the bucket angle θ becomes an angle threshold value θ_e (for example, about 30°) or more. It should be noted that the main controller 100 also sets the plunge flag to OFF when the excavation determination flag is set to OFF. The angle threshold value θ_e is a threshold value for determining whether or not the excavation work has been finished, and it is possible to employ, for example, the bucket angle θ when the crowding operation of the bucket 3 is completed and in a state where the bucket 3 is in the conveyance posture. That is, the excavation state determination section 132 determines that the excavation work has been finished when sensing the bucket 3 in the conveyance posture after it is determined that the excavation work has been started.

The torque distribution ratio calculation section 133 calculates the distribution ratio η_I [%] of the work driving torque and the distribution ratio η_C [%] of the travel driving torque on the basis of the reaction force F_R calculated by the reaction force calculation section 131, the excavation determination flag set by the excavation state determination section 132, and the torque distribution mode. The distribution ratio η_I of the work driving torque takes a value of 0 to 100%. Similarly, the distribution ratio η_C of the travel driving torque takes a value of 0 to 100%. The sum of the distribution ratio η_I of the work driving torque and the distribution ratio η_C of the travel driving torque is 100[%].

The torque distribution ratio calculation section 133, when the AUTO mode is set as the torque distribution mode, sets the distribution ratios η_I and η_C on the basis of the reaction force F_R calculated by the reaction force calculation section 131.

FIG. 6 is a diagram for showing a torque distribution ratio table for deciding the distribution ratio η_C of the travel

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driving torque and the distribution ratio η_I of the work driving torque. As shown by the solid line in FIG. 6, the torque distribution ratio table is a table in which the reaction force F_R received by the vehicle body 8 of the wheel loader 1 is associated with the distribution ratio η_C of the travel driving torque. It should be noted that the distribution ratio η_I of the work driving torque is indicated by the broken line in FIG. 6.

The torque distribution ratio table shown in FIG. 6 is stored in advance in the ROM 105 of the main controller 100. As shown by the solid line in FIG. 6, the torque distribution ratio table is a table having characteristics in which as the reaction force F_R becomes larger, the distribution ratio η_C of the travel driving torque becomes larger. In addition, the torque distribution ratio table is set such that as the reaction force F_R becomes larger, the ratio (inclination) of an increase in the distribution ratio η_C to an increase in the reaction force F_R becomes smaller. Accordingly, the travel driving force can be quickly increased according to an increase in the reaction force F_R when the wheel loader 1 is plunged into the excavation target 91. The torque distribution ratio table is defined in advance by experiments and the like.

The torque distribution ratio calculation section 133, when the excavation determination flag is set to ON, refers to the distribution ratio table of the travel driving torque shown by the solid line in FIG. 6 and calculates the distribution ratio η_C of the travel driving torque on the basis of the reaction force F_R calculated by the reaction force calculation section 131. In addition, the torque distribution ratio calculation section 133 subtracts the distribution ratio η_C [%] of the travel driving torque from 100[%] to calculate the distribution ratio η_I of the work driving torque ($100 - \eta_C = \eta_I$ [%]). Since the distribution ratio η_C of the travel driving torque becomes larger as the reaction force F_R becomes larger, the distribution ratio η_I of the work driving torque becomes smaller as the reaction force F_R becomes larger.

Therefore, when the AUTO mode is set as the torque distribution mode, the distribution ratios η_C and η_I change in real time according to the change in the reaction force F_R while the wheel loader 1 is performing the excavation work.

The torque distribution ratio calculation section 133, when the MANUAL mode is set as the torque distribution mode, sets setting values η_{CS} and η_{IS} of the distribution ratios stored in the ROM 105 as the distribution ratios η_C and η_I ($\eta_C = \eta_{CS}$ and $\eta_I = \eta_{IS}$). That is, when the MANUAL mode is set, the distribution ratios η_C and η_I are held at constant values even if the reaction force F_R changes.

The engine torque distribution calculation section 134 calculates a work driving torque command T_{I_COM} and a travel driving torque command T_{C_COM} on the basis of the distribution ratios η_C and η_I calculated by the torque distribution ratio calculation section 133, an engine output power torque TE , an auxiliary machine demanded torque T_{AUX_REQ} , a work demanded torque T_{I_REQ} , and a travel demanded torque T_{C_REQ} .

The engine output power torque TE , the auxiliary machine demanded torque T_{AUX_REQ} , the work demanded torque T_{I_REQ} , and the travel demanded torque T_{C_REQ} are calculated by the main controller 100. The engine output power torque TE is the maximum torque that can be output at the current engine revolution speed. The main controller 100 refers to an engine output power torque curve stored in the ROM 105 and calculates the engine output power torque TE on the basis of the engine revolution speed sensed by an engine revolution speed sensor. The auxiliary machine

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demanded torque T_{AUX_REQ} is calculated according to the operation states of a plurality of auxiliary machines operated by the electric power generated by the power generation motor 40. The main controller 100 sets a target value (for example, 1800 rpm) of the engine revolution speed. The target value of the engine revolution speed set by the main controller 100 is output to an engine controller that is not shown in the drawing. The engine controller controls a fuel injector (not shown) such that the engine revolution speed sensed by the engine revolution speed sensor becomes the target value.

The main controller 100 calculates the work demanded torque T_{I_REQ} on the basis of the operation amount of the arm operation lever 52 and the operation amount of the bucket operation lever 53. FIG. 7A is a diagram for showing an example of a pump demanded map used to calculate a pump demanded flow rate. The pump demanded map is stored in advance in the ROM 105 of the main controller 100. The main controller 100 refers to the pump demanded flow rate map shown in FIG. 7A and decides the pump demanded flow rate on the basis of the lever operation amount (lever signal). The pump demanded flow rate map is set such that the pump demanded flow rate is approximately proportional to the lever operation amount. The pump demanded flow rate becomes larger as the lever operation amount becomes larger. It should be noted that as the pump demanded flow rate maps, there are a map on the basis of the operation amount of the arm operation lever 52 and a map on the basis of the operation amount of the bucket operation lever 53, and the larger flow rate among those decided by the respective maps is decided as the pump demanded flow rate.

The main controller 100 computes hydraulic demanded power on the basis of the pump demanded flow rate and the delivery pressure of the hydraulic pump 30A sensed by the pressure sensor, and calculates the work demanded torque T_{I_REQ} on the basis of the hydraulic demanded power and the revolution speed of the engine 20 sensed by the revolution speed sensor. The work demanded torque T_{I_REQ} becomes larger as the lever operation amount becomes larger.

The main controller 100 calculates the travel demanded torque T_{C_REC} on the basis of the revolution speed of the travel motor 43 and the operation amount (accelerator signal) of the accelerator pedal 58. FIG. 7B is a diagram for showing an example of a torque map of the travel motor 43 used to calculate the travel demanded torque. The torque map is stored in advance in the ROM 105 of the main controller 100. A plurality of torque maps (torque curves) according to the accelerator signals are stored in the ROM 105 such that the torque of the travel motor 43 increases or decreases according to an increase or decrease in the accelerator signal. The torque map is set such that as the accelerator signal becomes larger, the travel demanded torque T_{C_REC} becomes larger, and as the revolution speed of the travel motor 43 becomes faster, the travel demanded torque T_{C_REC} becomes smaller.

The main controller 100 selects a torque map (torque curve) corresponding to the magnitude of the accelerator signal (the magnitude of the operation amount of the accelerator pedal 58) and decides the travel demanded torque T_{C_REC} on the basis of the revolution speed of the travel motor 43. For example, when the accelerator pedal 58 is fully operated (when the accelerator signal is maximum), the torque map of the solid line is selected, the selected torque map is referred to t_0 , and the travel demanded torque T_{C_REC} is calculated on the basis of the revolution speed of the travel motor 43. It should be noted that in a case where a trans-

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mission is provided, the main controller 100 calculates the travel demanded torque T_{C_REC} in consideration of the gear ratio of the transmission.

The engine torque distribution calculation section 134 shown in FIG. 4 decides a work target torque T_{I_TGT} on the basis of the work demanded torque T_{I_REQ} and the distribution ratio η_I of the work driving torque decided by the torque distribution ratio calculation section 133. The work target torque T_{I_TGT} is calculated by an equation (3).

[Equation 3]

$$T_{I_TGT} = T_{I_REQ} \eta_I \quad (3)$$

The engine torque distribution calculation section 134 decides a travel target torque T_{C_TGT} on the basis of the travel demanded torque T_{C_REC} and the distribution ratio η_C of the travel driving torque decided by the torque distribution ratio calculation section 133. The travel target torque T_{C_TGT} is calculated by an equation (4).

[Equation 4]

$$T_{C_TGT} = T_{C_REQ} \eta_C \quad (4)$$

The engine torque distribution calculation section 134 calculates a total target torque value T_{SUM_TGT} , which is the sum of the work target torque T_{I_TGT} , the travel target torque T_{C_TGT} , and the auxiliary machine demanded torque T_{AUX_REQ} ($T_{SUM_TGT} = T_{I_TGT} + T_{C_TGT} + T_{AUX_REQ}$). When the total target torque value T_{SUM_TGT} is equal to or less than the engine output power torque TE, the engine torque distribution calculation section 134 outputs the work target torque T_{I_TGT} as the work driving torque command T_{I_COM} and the travel target torque T_{C_TGT} as the travel driving torque command T_{C_COM} .

On the other hand, the engine torque distribution calculation section 134, when the total target torque value T_{SUM_TGT} is larger than the engine output power torque TE, decides the work driving torque command T_{I_COM} and the travel driving torque command T_{C_COM} such that a total command torque value T_{SUM_COM} , which is the sum of the work driving torque command T_{I_COM} , the travel driving torque command T_{C_COM} , and the auxiliary machine demanded torque T_{AUX_REQ} , does not exceed the engine output power torque TE. An example of the decision method will be described below.

The engine torque distribution calculation section 134 outputs a value obtained by subtracting a work driving torque correction value C_P , which is obtained by multiplying the amount $(T_{SUM_TGT} - TE)$ by which the total target torque value T_{SUM_TGT} exceeds the engine output power torque TE by the distribution ratio η_I , from the work target torque T_{I_TGT} as the work driving torque command T_{I_COM} . In addition, the engine torque distribution calculation section 134 outputs a value obtained by subtracting a travel driving torque correction value C_C , which is obtained by multiplying the amount $(T_{SUM_TGT} - TE)$ by which the total target torque value T_{SUM_TGT} exceeds the engine output power torque TE by the distribution ratio η_C , from the travel target torque T_{C_TGT} as the travel driving torque command T_{C_COM} . Accordingly, the work driving torque command T_{I_COM} and the travel driving torque command T_{C_COM} are decided such that the total command torque value T_{SUM_COM} does not exceed the engine output power torque TE without changing the ratios of the work driving torque command T_{I_COM} and the travel driving torque command T_{C_COM} .

It should be noted that the decision method of the work driving torque command T_{L_COM} and the travel driving torque command T_{C_COM} when the total target torque value T_{SUM_TGT} is larger than the engine output power torque TE is not limited to this. For example, the work driving torque command T_{L_COM} and the travel driving torque command T_{C_COM} may be decided such that the total command torque value T_{SUM_COM} does not exceed the engine output power torque TE by making a correction to subtract only one of the work target torque T_{L_TGT} and the travel target torque T_{C_TGT} .

The content of a torque decision process executed by the main controller 100 will be described with reference to FIG. 8. The process of the flowchart shown in FIG. 8 is started by turning on an ignition switch (engine key switch), and is repeatedly executed in a predetermined control cycle after performing the initial setting that is not shown in the drawing. It should be noted that the excavation determination flag is set to OFF in the initial setting.

As shown in FIG. 8, in Step S110, the main controller 100 calculates the reaction force F_R and proceeds to Step S120.

In Step S120, the main controller 100 performs a setting process of the excavation determination flag. When the bucket angle θ is not within the preset angle range (the lower threshold value θa to the upper threshold value θb), or when the bottom pressure Pa of the arm cylinder 4 is less than the preset pressure threshold value Pa0, the main controller 100 determines that the excavation work has not been started. In this case, the main controller 100 keeps the excavation determination flag OFF.

The main controller 100 determines that the excavation work has been started when the bucket angle θ is within the preset angle range (the lower threshold value θa to the upper threshold value θb), and when the bottom pressure Pa of the arm cylinder 4 is equal to or more than the preset pressure threshold value Pa0. In this case, the main controller 100 switches the excavation determination flag from OFF to ON.

When the setting process of the excavation determination flag is completed, the flow proceeds to Step S130. In Step S130, the main controller 100 determines whether the torque distribution mode is set to either the AUTO mode or the MANUAL mode. If it is determined in Step S130 that the torque distribution mode is set to the AUTO mode, the flow proceeds to Step S140, and if it is determined that the torque distribution mode is set to the MANUAL mode, the flow proceeds to Step S170.

In Step S170, the main controller 100 decides the setting value η_{CS} stored in the ROM 105 as the distribution ratio η_C and the setting value η_{S} as the distribution ratio η_S and proceeds to Step S180.

In Step S140, the main controller 100 determines whether or not the wheel loader 1 is in the state of performing the excavation work. When the excavation determination flag is set to ON in Step S140, it is determined that the wheel loader 1 is in the state of performing the excavation work, and the flow proceeds to Step S150. When the excavation determination flag is set to OFF in Step S140, it is determined that the wheel loader 1 is not in the state of performing the excavation work, and the flow proceeds to Step S160.

In Step S150, the main controller 100 decides the torque distribution ratios η_C and η_T on the basis of the reaction force F_R and proceeds to Step S180.

In Step S160, the main controller 100 decides a reference value η_{C0} (for example, 50%) as the distribution ratio η_C and a reference value η_{S0} (for example, 50%) as the

distribution ratio η_S and proceeds to Step S180. The reference values η_{C0} and η_{S0} are stored in advance in the ROM 105.

In Step S180, the main controller 100 generates the work driving torque command T_{L_COM} and the travel driving torque command T_{C_COM} on the basis of the distribution ratios η_C and η_T , the engine output power torque TE, the auxiliary machine demanded torque T_{AUX_REQ} , the work demanded torque T_{L_REQ} , and the travel demanded torque T_{C_REQ} , and terminates the process shown in the flowchart of FIG. 8.

The work driving torque command T_{L_COM} generated by the main controller 100 is output to a pump controller that is not shown in the drawing. The pump controller generates a control signal for controlling the delivery capacity (displacement volume) of the hydraulic pump 30A on the basis of the work driving torque command T_{L_COM} and the delivery pressure of the hydraulic pump 30A. The pump controller controls the delivery capacity of the hydraulic pump 30A by outputting the generated control signal to a regulator that is not shown in the drawing. Accordingly, the work device 6 (the arms 2 and the bucket 3) is driven by the work driving force generated by the hydraulic cylinders 4 and 5. As described above, the main controller 100 according to the embodiment controls the work driving torque by computing the reaction force F_R acting on the wheel loader 1, deciding the distribution ratio η_T of the work driving torque on the basis of the reaction force F_R , and generating the work driving torque command T_{L_COM} on the basis of the distribution ratio η_T .

The travel driving torque command T_{C_COM} generated by the main controller 100 is output to the travel inverter 42. The travel inverter 42 drives the travel motor 43 on the basis of the travel driving torque command T_{C_COM} . The torque generated by the travel motor 43 is transmitted to the tires 7 configuring a part of the travel device 11 via the power transmission device configuring a part of the travel device 11. Accordingly, the travel device 11 is driven by the travel driving force generated by the travel motor 43. As described above, the main controller 100 according to the embodiment controls the travel driving torque by computing the reaction force F_R acting on the wheel loader 1, deciding the distribution ratio η_C of the travel driving torque on the basis of the reaction force F_R , and generating the travel driving torque command T_{C_COM} on the basis of the distribution ratio η_C .

With reference to FIG. 9, the travel driving torque command and the work driving torque command when the excavation work is performed by the wheel loader 1 according to the embodiment will be described. FIG. 9 is time charts for showing the operation of the main controller 100 according to the embodiment. In order to clarify working effects of the embodiment, an explanation will be made while comparing with the operation of a main controller of a wheel loader according to a comparative example of the embodiment. It should be noted that the wheel loader according to the comparative example has a configuration in which the AUTO mode is not provided, the distribution ratio η_C is set at 50%, the distribution ratio η_T is set at 50%, and the distribution ratios η_C and η_T do not change during the excavation work. In the drawings, the operation of the main controller 100 according to the embodiment is shown by a solid line, and the operation of the main controller according to the comparative example is shown by a broken line.

The horizontal axis in FIG. 9 shows time (elapsed time). The vertical axis in FIG. 9(a) shows the reaction force F_R calculated by the main controller 100, the vertical axis in FIG. 9(b) shows the distribution ratio η_C of the travel driving

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torque decided by the main controller **100**, the vertical axis in FIG. 9(c) shows the travel driving torque command $T_{C.COM}$ generated by the main controller **100**, and the vertical axis in FIG. 9(d) shows the work driving torque command $T_{L.COM}$ generated by the main controller **100**.

It should be noted that in both the embodiment and the comparative example, it is assumed that the operation procedure and the operation amount by the operator for various operation members are the same. A time T_0 is the time when the wheel loader **1** plunged into the excavation target **91** such as a sediment pile (that is, the time when the penetration of the bucket **3** into the excavation target **91** has been started). That is, the time T_0 is the time when the plunge flag has been set to ON. A time T_1 is the time when the excavation determination flag has been set to ON, and a time T_2 is the time when the excavation determination flag has been set to OFF. A time T_a is the time when the operator has started operating the arm operation lever **52**. In addition, it is assumed that the operator has switched the forward/backward switch **51** from the forward position to the backward position at the time T_2 .

As shown in FIG. 9(a), the reaction force F_R is small until the time T_0 . This is because since the wheel loader **1** is not brought into contact with the excavation target **91** until the time T_0 , only the reaction force of the ground (travel surface) during travelling is calculated. At the time T_0 , since the wheel loader **1** is brought into contact with the excavation target **91**, the repulsive force from the excavation target **91** is added. Therefore, the reaction force F_R sharply rises from the time T_0 . It should be noted that in the embodiment, the reaction force is sequentially calculated with the time T_0 as the reference time t_0 . For example, a reaction force F_{Rb} at a time T_b is calculated by the above equation (1) on the basis of the mass m of the wheel loader **1**, the elapsed time $(t_1 - t_0)$ from the time T_0 (reference time t_0), the vehicle velocity v_0 at the time T_0 , and the vehicle velocity v_1 and the traction force F_P at the time T_b .

In the comparative example, the distribution ratios η_C and η_L of the torque are not decided on the basis of the reaction force F_R . Therefore, in the comparative example, the distribution ratio η_C of the travel driving torque always becomes a constant value as shown in FIG. 9(b). Therefore, as shown in FIG. 9(c), the travel driving torque command also always becomes a constant value.

On the other hand, in the embodiment, as the reaction force F_R becomes larger, the distribution ratio η_C of the travel driving torque becomes larger between the time when the excavation determination flag is set to ON and the time when it is set to OFF (from the time T_1 to the time T_2), that is, during the time when the main controller **100** determines that the excavation work is performed.

Accordingly, as the reaction force F_R becomes larger, the travel driving torque command becomes larger from the time T_1 to the time T_2 . It should be noted that in other periods, the distribution ratio η_C and the travel driving torque command of the embodiment are the same as those of the comparative example.

Although not shown in the drawing, the distribution ratio η_L of the work driving torque always becomes a constant value in the comparative example. Therefore, as shown in FIG. 9(d), when the operation of the arm operation lever **52** is started (the time T_a) and the arm operation lever **52** is raised up to a predetermined operation amount (for example, the maximum operation amount), the work driving torque command becomes a constant value in the comparative example. On the other hand, in the embodiment, when the arm operation lever **52** is raised up to a predetermined

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operation amount (for example, the maximum operation amount) and held at that operation amount, the work driving torque command becomes smaller as the reaction force F_R becomes larger. It should be noted that after the time T_2 , the distribution ratio η returns to the value before the time T_1 in the embodiment. Therefore, after the time T_2 , the work driving torque command in the embodiment becomes the same value as the work driving torque command in the comparative example.

As described above, in the embodiment, the main controller **100** controls the output power of the hydraulic pump **30A** and the output power of the travel motor **43** by changing the distribution ratios η_C and η_L on the basis of the reaction force F_R acting on the vehicle body **8** of the wheel loader **1** during the excavation work to control the work driving torque and the travel driving torque. Therefore, the work efficiency can be improved because the distribution ratios of the travel driving force and the work driving force can be made closer to a distribution ratio suitable for the hardness of the excavation target **91** without depending on the skill level of the operator. In addition, in the embodiment, since the distribution ratios η_C and η_L are sequentially adjusted, the work efficiency can be improved even in a case where the excavation target **91** contains a plurality of soils with different degrees of hardness. Further, the reaction force F_R acting on the vehicle body **8** of the wheel loader **1** is affected by road surface conditions. Therefore, according to the embodiment, even in a case where the hardness of the excavation target **91** and the road surface conditions change in each excavation cycle or change in real time during the excavation work, the work efficiency can be improved because the torque distribution ratio can be made closer to an appropriate value.

According to the above-described embodiment, the following working effects can be obtained.

(1) A wheel loader (work vehicle) **1** includes: an engine **20** that is mounted on a vehicle body **8**; a hydraulic pump **30A** that is driven by the engine **20**; hydraulic cylinders **4** and **5** that are extended and contracted by pressurized fluid delivered from the hydraulic pump **30A**; a work device **6** that is moved according to the extension/contraction operations of the hydraulic cylinders **4** and **5**; a travel device **11** that is driven independently of the work device **6**; a travel motor (electrically driven motor) **43** that is driven by electric power generated by the engine **20** to operate the travel device **11**; and a main controller (controller) **100** that controls the hydraulic cylinders **4** and **5** and the travel motor **43**. The controller **100** controls the output power of the hydraulic pump **30A** and the output power of the travel motor **43** by changing the distribution ratios η_L and η_C of a work driving torque (first torque) consumed by the work device **6** and a travel driving torque (second torque) consumed by the travel device **11** among torques output by the engine **20**, on the basis of a reaction force F_R received by the vehicle body **8**.

Therefore, in a case where the excavation work is repeatedly performed, even when excavating the excavation target **91** with a different degree of hardness, the power of the engine **20** can be appropriately distributed to the work device **6** and the travel device **11**. That is, according to the embodiment, the efficiency of the excavation work for the excavation target **91** with a different degree of hardness can be improved.

(2) In addition, even if the skill level of the operator is low and the operation amounts of the arm operation lever **52**, the bucket operation lever **53**, and the accelerator pedal **58** deviate from appropriate operation amounts, the distribution

of the work driving torque and the travel driving torque can be made closer to appropriate distribution. Accordingly, the efficiency of the excavation work can be improved regardless of the skill level of the operator.

(3) The main controller **100** determines whether or not the wheel loader **1** is in the state of performing the excavation work, and while the wheel loader **1** is determined to be in the state of performing the excavation work (that is, while the wheel loader **1** is performing the excavation work), the work driving torque and the travel driving torque are controlled such that as the reaction force F_R becomes larger, the work driving torque becomes smaller and the travel driving torque becomes larger. Accordingly, when the hardness of the excavation target **91** changes in real time, the travel driving torque and the work driving torque can be appropriately set. In addition, according to this configuration, even when the road surface conditions change in real time, the travel driving torque and the work driving torque can be appropriately set.

(4) The wheel loader **1** includes the mode switching switch (mode switching operation section) **57** that can be manually switched between the AUTO mode (first control mode) in which the work driving torque and the travel driving torque are controlled on the basis of the reaction force F_R and the MANUAL mode (second control mode) in which the work driving torque and the travel driving torque are controlled such that the ratios of the work driving torque and the travel driving torque are held at predetermined ratios (for example, the setting values η_{TS} and η_{CS}) regardless of the reaction force F_R . Therefore, the operator can properly use the AUTO mode and the MANUAL mode depending on the condition.

Second Embodiment

A wheel loader **1** according to a second embodiment will be described mainly with reference to FIG. **10**. It should be noted that the same or equivalent parts as those in the first embodiment are given the same reference numerals in the drawings, and the differences are mainly explained. FIG. **10** is diagrams similar to FIG. **9** and is time charts for showing the operation of a main controller **100** according to the second embodiment.

The wheel loader **1** according to the second embodiment has a configuration similar to the wheel loader **1** according to the first embodiment, but processes in the torque distribution ratio calculation section **133** are different from the first embodiment. As shown in FIG. **10**, the torque distribution ratio calculation section **133** decides the distribution ratio η_T of the work driving torque and the distribution ratio η_C of the travel driving torque on the basis of a reaction force F_{RC} calculated at a time T_c after the lapse of a predetermined time ΔT_p from the time T_0 when the plunge flag is set to ON. The torque distribution ratios η_T and η_C are held until the excavation determination flag is switched from ON to OFF. When the excavation determination flag is set from ON to OFF, the controller **100** initializes the distribution ratios η_T and η_C to the reference values η_{T0} and η_{C0} .

The predetermined time ΔT_p can be appropriately set according to the work content and performance of the wheel loader **1**, the calculation ability of the main controller **100**, and the like. In order to increase calculation accuracy, a long predetermined time ΔT_p is better, and by securing, for example, about 0.1 seconds, the reaction force F_{RC} can be calculated with a certain degree of calculation accuracy. In addition, the time (the time T_0 to the time T_2) from when the wheel loader **1** plunges into the excavation target **91** to the

end of the excavation work is approximately 5 seconds. Therefore, it is preferable that the value of the predetermined time ΔT_p is 0.1 seconds or more and 5 seconds or less.

In addition, the time from plunging the wheel loader **1** into the excavation target **91** to starting to raise the arms **2** is approximately 0.5 seconds, and the time from plunging the wheel loader **1** into the excavation target **91** to starting the crowding operation of the bucket **3** is approximately 1.5 seconds. Therefore, it is more preferable to set the predetermined time ΔT_p in the range of 0.5 seconds or more and 1.5 seconds or less. In particular, by setting the predetermined time ΔT_p to about 0.5 seconds, the torque distribution ratio can be quickly fixed and reflected in the work driving torque command and the travel driving torque command.

The torque distribution ratio calculation section **133** holds the distribution ratios η_T and η_C calculated on the basis of the reaction force F_{RC} calculated at the time T_c until the excavation determination flag is switched from ON to OFF. Accordingly, even if the reaction force acting on the wheel loader **1** temporarily becomes smaller than the reaction force F_{RC} by releasing the accelerator pedal **58** or moving back the vehicle body during the excavation work, the distribution ratios η_T and η_C of the torque do not fluctuate. Accordingly, in a case where the wheel loader **1** is moved forward and backward a plurality of times to scoop up the sediment in a single excavation work, appropriate torque distribution ratios η_T and η_C can be set in advance during the second and subsequent scooping up work of the sediment.

As described above, the main controller **100** according to the second embodiment controls the work driving torque and the travel driving torque such that it is determined whether or not the wheel loader **1** has plunged into the excavation target **91**, the distribution ratios η_T and η_C of the work driving torque and the travel driving torque are decided on the basis of the reaction force F_{RC} after the elapse of the predetermined time ΔT_p from the time when being determined that the wheel loader **1** has plunged into the excavation target **91** (from the time when the plunge flag has been set to ON), and the distribution ratios η_T and η_C are held until the excavation work is finished (until the excavation determination flag is set from ON to OFF).

According to such a second embodiment, the excavation work can be performed with more work patterns. That is, according to the second embodiment, in addition to the working effects similar to the first embodiment, the degree of freedom of the excavation work can be improved.

Third Embodiment

A wheel loader **1** according to a third embodiment will be described mainly with reference to FIG. **11** and FIG. **12**. It should be noted that the same or equivalent parts as those in the first embodiment are given the same reference numerals in the drawings, and the differences are mainly explained. FIG. **11** is a diagram similar to FIG. **4** and is a functional block diagram of a main controller **300** according to the third embodiment.

As shown in FIG. **2**, in the hydraulic fluid line connecting the bottom-side hydraulic fluid chamber (not shown) of the arm cylinder **4** and the front device control section **31** to each other, a relief valve **72** is provided to regulate the maximum pressure of the bottom pressure P_a of the arm cylinder **4** to a preset relief pressure. If the bottom pressure P_a of the arm cylinder **4** rises to the relief pressure during the process in which the wheel loader **1** plunges into the excavation target **91** and the bucket **3** is penetrated into the excavation target **91**, there is a risk that the raising operation

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of the arms **2** is not appropriately performed. Thus, in the third embodiment, the raising operation of the arms **2** is appropriately performed by deciding the distribution ratios η_I and η_C of the torque in consideration of the bottom pressure Pa of the arm cylinder **4**.

As shown in FIG. **11**, the torque distribution ratio calculation section **333**, when the excavation determination flag is set to ON, calculates the distribution ratios η_I and η_C of the torque on the basis of the reaction force F_R and the bottom pressure Pa of the arm cylinder **4**.

FIG. **12** is a diagram similar to FIG. **6** and showing a torque distribution ratio table according to the third embodiment. As shown in FIG. **12**, a plurality of torque distribution ratio tables according to the magnitude of the bottom pressure Pa of the arm cylinder **4** are stored in the ROM **105** such that the distribution ratio η_C of the travel driving torque changes according to the magnitude of the bottom pressure Pa of the arm cylinder **4**.

The main controller **300** selects a torque distribution ratio table corresponding to the magnitude of the bottom pressure Pa of the arm cylinder **4** and decides the distribution ratio η_C of the travel driving torque on the basis of the reaction force F_R . The torque distribution ratio table is set such that as the reaction force F_R becomes larger, the distribution ratio η_C of the travel driving torque becomes larger and as the bottom pressure Pa of the arm cylinder **4** becomes higher, the distribution ratio η_C of the travel driving torque becomes smaller.

Therefore, in the wheel loader **1** according to the third embodiment, since the bottom pressure Pa of the arm cylinder **4** is small at the beginning of the penetration of the bucket **3** into the excavation target **91**, the distribution ratio η_C of the travel driving torque becomes large. Then, since the bottom pressure Pa of the arm cylinder **4** becomes large from the middle to the end of the penetration of the bucket **3** into the excavation target **91**, the distribution ratio η_C of the travel driving torque becomes small.

As described above, the main controller **300** according to the third embodiment controls the travel driving torque such that as the pressure of the arm cylinder (hydraulic cylinder) **4** for operating the arms **2** becomes larger, the travel driving torque becomes smaller. Since the bottom pressure Pa of the arm cylinder **4** can be lowered by making the travel driving torque smaller as the bottom pressure Pa of the arm cylinder **4** becomes closer to the relief pressure, the lifting operation of the arms **2** can be appropriately performed.

According to such a third embodiment, when the sediment put in the bucket **3** is heavy or the arm raising operation is delayed, the raising operation of the arms **2** (lifting of the bucket **3**) being inoperable due to reaching of the bottom pressure Pa of the arm cylinder **4** to the maximum pressure (relief pressure) by an increase in the travel driving force can be prevented or delayed. Therefore, according to the third embodiment, in addition to the working effects similar to the first embodiment, it is possible to suppress a decrease in the efficiency of the excavation work from the middle to the end of the penetration of the bucket **3** into the excavation target **91**.

It should be noted that, in the embodiment, the method of deciding the distribution ratio η_C of the travel driving torque by selecting a table according to the bottom pressure Pa of the arm cylinder **4** from among the plurality of torque distribution ratio tables has been described, but the method of deciding the distribution ratio η_C is not limited to this. For example, a reference distribution ratio is decided on the basis of the torque distribution ratio table described in the first embodiment, and the distribution ratio η_C may be

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decided by multiplying the reference distribution ratio by a factor preset according to the bottom pressure Pa of the arm cylinder **4**.

Fourth Embodiment

A wheel loader **1** according to a fourth embodiment will be described mainly with reference to FIG. **13**. It should be noted that the same or equivalent parts as those in the first embodiment are given the same reference numerals in the drawing, and the differences are mainly explained. FIG. **13** is a diagram similar to FIG. **4** and is a functional block diagram of a main controller **400** according to the fourth embodiment.

The main controller **400** according to the fourth embodiment further has a function as a torque upper limit value calculation section **433** for deciding the upper limit value T_{I_LIM} of the work driving torque and the upper limit value T_{C_LIM} of the travel driving torque.

The torque upper limit value calculation section **433** calculates a difference value ΔTE , which is the value obtained by subtracting the auxiliary machine demanded torque T_{AUX_REQ} from the engine output power torque TE ($\Delta TE = TE - T_{AUX_REQ}$).

The torque upper limit value calculation section **433** calculates a work driving torque upper limit value T_{I_LIM} by multiplying the distribution ratio η_I of the work driving torque decided by the torque distribution ratio calculation section **133** by the difference value ΔTE ($T_{I_LIM} = \Delta TE \cdot \eta_I$).

The torque upper limit value calculation section **433** calculates a travel driving torque upper limit value T_{C_LIM} by multiplying the distribution ratio η_C of the travel driving torque decided by the torque distribution ratio calculation section **133** by the difference value ΔTE ($T_{C_LIM} = \Delta TE \cdot \eta_C$).

The engine torque distribution calculation section **434** decides the work target torque T_{I_TGT} on the basis of the work demanded torque T_{I_REQ} and the work driving torque upper limit value T_{I_LIM} decided by the torque upper limit value calculation section **433**. The work target torque T_{I_TGT} is calculated by an equation (5).

[Equation 5]

$$T_{I_TGT} = \begin{cases} T_{I_REQ} & (T_{I_REQ} \leq T_{I_LIM}) \\ T_{I_LIM} & (T_{I_REQ} > T_{I_LIM}) \end{cases} \quad (5)$$

The engine torque distribution calculation section **434** decides the travel target torque T_{C_TGT} on the basis of the travel demanded torque T_{C_REQ} and the travel driving torque upper limit value T_{C_LIM} decided by the torque upper limit value calculation section **433**. The travel target torque T_{C_TGT} is calculated by an equation (6).

[Equation 6]

$$T_{C_TGT} = \begin{cases} T_{C_REQ} & (T_{C_REQ} \leq T_{C_LIM}) \\ T_{C_LIM} & (T_{C_REQ} > T_{C_LIM}) \end{cases} \quad (6)$$

As described above, the main controller **400** according to the fourth embodiment decides the upper limit value T_{I_LIM} of the work driving torque and the upper limit value T_{C_LIM} of the travel driving torque on the basis of the reaction force F_R , and controls the work driving torque and the travel driving torque on the basis of the upper limit value T_{I_LIM} of

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the work driving torque and the upper limit value T_{C_LIM} of the travel driving torque. That is, the main controller **400** can change the distribution ratios of the work driving torque and the travel driving torque on the basis of the reaction force F_R by controlling the work driving torque and the travel driving torque on the basis of the work driving torque upper limit value T_{I_LIM} and the travel driving torque upper limit value T_{C_LIM} .

In the fourth embodiment, when the work demanded torque and the travel demanded torque exceed the upper limit values (T_{I_LIM} and T_{C_LIM}) during the excavation work, the work driving torque and the travel driving torque are controlled such that the distribution ratios of the work driving torque and the travel driving torque become close to proper distribution ratios. Accordingly, the work driving torque and the travel driving torque intended by the operator can be generated until the upper limit values are exceeded. For operators who feel such a configuration easier to handle, the work efficiency can be improved over the first embodiment.

In addition, in the fourth embodiment, the travel demanded torque reaches the upper limit value of the travel driving torque when the operation amount of the accelerator pedal **58** is smaller than the maximum operation amount, or the work demanded torque reaches the upper limit value of the work driving torque when the operation amounts of the arm operation lever **52** and the bucket operation lever **53** are smaller than the maximum operation amounts. Therefore, it is possible to reduce the frequency at which the operator steps on the accelerator pedal **58** to maximize the operation amount of the accelerator pedal **58** or the operator tilts the arm operation lever **52** and the bucket operation lever **53** until the operation amounts of the arm operation lever **52** and the bucket operation lever **53** are maximized. As a result, the total operation amount by the operator can be decreased, and the burden on the operator can be reduced.

Fifth Embodiment

A wheel loader **1** according to a fifth embodiment will be described mainly with reference to FIG. **14**. It should be noted that the same or equivalent parts as those in the first embodiment are given the same reference numerals in the drawings, and the differences are mainly explained. FIG. **14** is a diagram similar to FIG. **4** and is a functional block diagram of a main controller **500** according to the fifth embodiment.

The main controller **500** according to the fifth embodiment has, instead of the torque distribution ratio calculation section **133** of the main controller **100** according to the first embodiment, a function as a torque upper limit value calculation section **533** for deciding the upper limit value T_{I_LIM} of the work driving torque and the upper limit value T_{C_LIM} of the travel driving torque.

The torque upper limit value calculation section **533** computes the torque upper limit values T_{I_LIM} and T_{C_LIM} on the basis of the reaction force F_R . For example, the torque upper limit value calculation section **533** refers to a preset upper limit value table, and computes the torque upper limit values T_{I_LIM} and T_{C_LIM} on the basis of the reaction force F_R . The upper limit value table is defined in advance by experiments and the like and stored in the ROM **105**.

The upper limit value table of the travel driving torque is a table having characteristics in which as the reaction force F_R becomes larger, the travel driving torque upper limit value T_{C_LIM} becomes larger. In addition, the upper limit value table of the travel driving torque is set such that as the

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reaction force F_R becomes larger, the ratio (inclination) of an increase in the travel driving torque upper limit value T_{C_LIM} an increase in the reaction force F_R becomes smaller. The upper limit value table of the work driving torque is a table having characteristics in which as the reaction force F_R becomes larger, the work driving torque upper limit value T_{I_LIM} becomes smaller. In addition, the upper limit value table of the work driving torque is set such that as the reaction force F_R becomes larger, the ratio (inclination) of an increase in the work driving torque upper limit value T_{I_LIM} to an increase in the reaction force F_R becomes larger.

The engine torque distribution calculation section **534** decides the work target torque T_{I_TGT} by using the above equation (5) on the basis of the work demanded torque T_{I_REQ} and the work driving torque upper limit value T_{I_LIM} decided by the torque upper limit value calculation section **533**.

The engine torque distribution calculation section **534** decides the travel target torque T_{C_TGT} by using the above equation (6) on the basis of the travel demanded torque T_{C_REQ} and the travel driving torque upper limit value T_{C_LIM} decided by the torque upper limit value calculation section **533**.

As described above, the main controller **500** according to the fifth embodiment can change the distribution ratios of the work driving torque and the travel driving torque on the basis of the reaction force F_R by controlling the work driving torque and the travel driving torque on the basis of the work driving torque upper limit value T_{I_LIM} and the travel driving torque upper limit value T_{C_LIM} . According to such a fifth embodiment, the working effects similar to the fourth embodiment can be obtained.

The following modified examples are also within the scope of the present invention, and it is possible to combine the configurations shown in the modified examples with the configurations described in the above-described embodiments, to combine the configurations described in the above-described different embodiments with each other, or to combine the configurations described in the following different modified examples with each other.

Modified Example 1

The calculation method of the work driving torque command T_{I_COM} and the travel driving torque command T_{C_COM} is not limited to the method described in the above embodiments. As a modified example of the first embodiment, the engine torque distribution calculation section **134** may calculate the work driving torque command T_{I_COM} and the travel driving torque command T_{C_COM} on the basis of the distribution ratios η_C and η_I calculated by the torque distribution ratio calculation section **133**, the engine output power torque TE , the auxiliary machine demanded torque T_{AUX_REQ} , the work demanded torque T_{I_REQ} , and the travel demanded torque T_{C_REQ} , as will be described below.

The engine torque distribution calculation section **134** calculates a total demanded torque value T_{SUM_REQ} , which is the sum of the work demanded torque T_{I_REQ} , the travel demanded torque T_{C_REQ} , and the auxiliary machine demanded torque T_{AUX_REQ} ($T_{SUM_REQ} = T_{I_REQ} + T_{C_REQ} + T_{AUX_REQ}$).

The engine torque distribution calculation section **134**, when the total demanded torque value T_{SUM_REQ} is equal to or less than the engine output power torque TE , sets the work demanded torque T_{I_REQ} as the work target torque T_{I_TGT} and the travel demanded torque T_{C_REQ} as the travel target torque T_{C_TGT} .

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The engine torque distribution calculation section 134, when the total demanded torque value T_{SUM_REQ} is larger than the engine output power torque TE, calculates the work target torque T_{L_TGT} and the travel target torque T_{C_TGT} by subtracting a predetermined amount from the work demanded torque T_{L_REQ} and the travel demanded torque T_{C_REQ} in accordance with the torque distribution ratio η by the difference value ΔTE between the total demanded torque value T_{SUM_REQ} and the engine output power torque TE. In this case, the engine torque distribution calculation section 134 makes the total target torque value T_{SUM_TGT} , which is the sum of the auxiliary machine demanded torque T_{AUX_REQ} , the work target torque T_{L_TGT} , and the travel target torque T_{C_TGT} , equal to the engine output power torque TE.

According to such a modified example, the maximum values of the travel driving force and the work driving force are not reduced even when the excavation determination flag is set to ON. Therefore, the wheel loader 1 can be operated as more intended by the operator while improving the work efficiency. According to the modified example, the work efficiency can be improved as similar to the first embodiment, and the operability of the vehicle body can be further improved.

Modified Example 2

As a modified example of the first embodiment, the engine torque distribution calculation section 134 may calculate the work target torque T_{L_TGT} by an equation (7) and the travel target torque T_{C_TGT} by an equation (8).

[Equation 7]

$$T_{L_TGT} = T_{L_REQ} \cdot \eta_l \cdot k \quad (7)$$

[Equation 8]

$$T_{C_TGT} = T_{C_REQ} \cdot \eta_c \cdot k \quad (8)$$

In the equation, k is a constant and stored in advance in the ROM 105. For example, when k is 1, the work target torque T_{L_TGT} becomes the value equal to or less than the work demanded torque T_{L_REQ} , and the travel target torque T_{C_TGT} becomes the value equal to or less than the travel demanded torque T_{C_REQ} . For example, when k is set to 1, when η_l is 50% and η_c is 50%, the work target torque T_{L_TGT} becomes the value half the work demanded torque T_{L_REQ} , and the travel target torque T_{C_TGT} becomes the value half the travel demanded torque T_{C_REQ} . On the other hand, when k is larger than 1, the work target torque T_{L_TGT} and the travel target torque T_{C_TGT} can be made larger, and thus it is possible to be made closer to the work driving force and the travel driving force intended by the operator. For example, when k is set to 2, when η_l is 50% and η_c is 50%, the work target torque T_{L_TGT} becomes the value of the work demanded torque T_{L_REQ} , and the travel target torque T_{C_TGT} becomes the value of the travel demanded torque T_{C_REQ} . It should be noted that the value of k may be optionally set by the operator by manual operation.

Modified Example 3

In the second embodiment described with reference to FIG. 10, an example of deciding the distribution ratio η_l of the work driving torque and the distribution ratio η_c of the travel driving torque on the basis of the reaction force F_{RC} calculated at the time Tc after the lapse of the predetermined time ΔTp from the time T0 when the plunge flag is set to ON

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has been described. As a modified example, the reaction force F_R is sequentially calculated after the plunge flag is set to ON, and when the reaction force F_R decreases after the excavation determination flag is set to ON, the distribution ratio η_l of the work driving torque and the distribution ratio η_c of the travel driving torque decided on the basis of the reaction force F_R at that time may be held until the excavation determination flag is set to OFF. Accordingly, as similar to the second embodiment, in a case where the wheel loader 1 is moved forward and backward a plurality of times to scoop up the sediment in a single excavation work, appropriate torque distribution ratios η_l and η_c can be set in advance during the second and subsequent scooping up work of the sediment.

Modified Example 4

In the above embodiments, an example in which the power generation inverter 41 and the travel inverter 42 are connected to each other via the DC section 44 has been described, but the present invention is not limited to this. The power generation inverter 41 and the travel inverter 42 may be power converters without via a DC section as represented by a matrix converter.

Modified Example 5

In the above embodiments, an example of the wheel loader 1 without a power storage device connected to the DC section 44 has been described, but the present invention is not limited to this. The present invention can also be applied to a wheel loader having a configuration in which the voltage of the DC section 44 is controlled or electric power is supplied by connecting a power storage device including a power storage element such as a secondary battery or a capacitor to the DC section 44.

Modified Example 6

In the above embodiments, an example of calculating the traction force F_P on the basis of the equation (2) has been described, but the present invention is not limited to this. For example, the bottom pressure Pa of the arm cylinder 4 is proportional to the traction force F_P . Therefore, the traction force F_P may be calculated on the basis of the bottom pressure Pa of the arm cylinder 4 sensed by the pressure sensor 71. In addition, an acceleration sensor is attached to the wheel loader 1, and the traction force F_P may be calculated on the basis of the acceleration sensed by the acceleration sensor and the mass of the wheel loader 1.

Modified Example 7

The method of determining that the excavation work has been started is not limited to the method described in the above embodiments. For example, it may be determined that the excavation work has been started in consideration of the reaction force F_R calculated by the reaction force calculation section 131. For example, the excavation state determination section 132 determines that the excavation work has been started when the bucket 3 is within the angle range indicating that the bucket 3 is in the plunging posture, the bottom pressure Pa of the arm cylinder 4 is equal to or more than the pressure threshold value Pa0, and the reaction force F_R is equal to or more than a predetermined reaction force threshold value. Accordingly, it is possible to reduce the occurrence of erroneous determination in which it is determined

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as the excavation state when the excavation work is not being performed. In addition, instead of the determination method described in the above embodiments, it may be determined whether or not the excavation work has been started on the basis of image data photographed by a camera (photographing device) for monitoring the front of the wheel loader **1**. It may be determined whether or not the excavation work has been started on the basis of information sensed by an infrared sensor for monitoring the front of wheel loader **1**.

Modified Example 8

In the above embodiments, an example in which after it is determined that the excavation work has been started, it is determined that the excavation work has been finished when the bucket angle θ becomes the angle threshold value θ_e or larger has been described, but the present invention is not limited to this. For example, it may be determined that the excavation work has been finished when the time after being determined that the excavation work has been started has passed a preset time threshold value (for example, about 5 seconds). The time threshold value is stored in advance in the ROM **105**. The time threshold value is defined in advance by experiments and the like.

Modified Example 9

In the first embodiment, an example in which as shown in FIG. **8**, the main controller **100** decides the distribution ratio η_I of the work driving torque and the distribution ratio η_C of the travel driving torque on the basis of the reference values η_I0 and η_C0 in Step **S160**, and decides the distribution ratio η_I of the work driving torque and the distribution ratio η_C of the travel driving torque on the basis of the setting values η_{IS} and η_{CS} set by the torque distribution ratio setting dial **54** in Step **S170** has been described.

However, in Step **S160**, the main controller **100** may decide the distribution ratio η_I of the work driving torque and the distribution ratio η_C of the travel driving torque on the basis of the setting values η_{IS} and η_{CS} set by the torque distribution ratio setting dial **54**. In addition, in Step **S170**, the main controller **100** may decide the distribution ratio η_I of the work driving torque and the distribution ratio η_C of the travel driving torque on the basis of the reference values η_I0 and η_C0 . It should be noted that in both Steps **S160** and **S170**, in a case where the distribution ratios η_I and η_C are decided on the basis of the reference values η_I0 and η_C0 , the torque distribution ratio setting dial **54** can be omitted.

Modified Example 10

In the above embodiments, an example in which the main controllers **100**, **300**, **400**, and **500** calculate the reaction force F_R acting on the vehicle body **8** of the wheel loader **1** and decides the torque distribution ratio and the torque upper limit value on the basis of the calculated reaction force F_R has been described, but the present invention is not limited to this. The main controllers may calculate the physical quantity correlated with the reaction force F_R acting on the wheel loader **1** and decide the torque distribution ratio and the torque upper limit value on the basis of the calculated physical quantity. As the physical quantity correlated with the reaction force F_R , for example, the bottom pressure P_a of the arm cylinder **4** can be used. In a case where the torque distribution ratio and the torque upper limit value are decided on the basis of the physical quantity correlated with

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the reaction force F_R as described above, it can be said that the main controllers control the work driving torque and the travel driving torque on the basis of the reaction force received by the vehicle body **8** of the wheel loader **1**. Accordingly, it is possible to set the torque distribution ratio during the excavation work on a work site where the ground is inclined from the horizontal plane in the same manner as the torque distribution ratio during the excavation work on a work site where the ground is parallel to the horizontal plane. That is, the effect of the inclination of the ground can be reduced.

Modified Example 11

In the first embodiment, an example of sequentially calculating the reaction force F_R on the basis of the vehicle velocity v of the wheel loader **1** has been described, but the calculation method of the reaction force F_R is not limited to this. The time change ratio of the vehicle velocity v at the beginning of penetration into the excavation target **91** is compared with the time change ratio of the vehicle velocity v stored in advance in the ROM **105**, and when the both match each other, the reaction force F_R may be decided from a reaction force calculation data table stored in the ROM **105**. The reaction force calculation data table is defined in advance for each time change ratio of a plurality of vehicle velocities v (that is, for each of a plurality of vehicle velocity change patterns) by experiments and the like.

Modified Example 12

In the above embodiments, the values used for various determinations and calculations may be subjected to moving average processing or low-pass filter processing in order to avoid the effects of disturbance and noise.

Modified Example 13

Some or all of the functions of the main controllers described in the above embodiments may be realized by hardware (for example, by designing the logic to execute each function using an integrated circuit).

Although the embodiments of the present invention have been described above, the above embodiments show only a part of applications of the present invention, and the technical scope of the present invention is not intended to be limited to the specific configuration of the above embodiments. The above-described embodiments and modified examples are exemplified for the purpose of clearly describing the present invention and are not necessarily limited to those having all the described configurations. In addition, it is possible to replace a part of the configuration of one embodiment or modified example with the configuration of another embodiment or modified example, and to add the configuration of one embodiment or modified example to the configuration of another embodiment or modified example. It should be noted that the control lines and the information lines considered to be necessary in the explanation are shown in the drawings, but all the necessary control lines and information lines in a product are not necessarily shown. In practice, almost all the configurations may be considered to be connected to each other.

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DESCRIPTION OF REFERENCE CHARACTERS

1: Wheel loader (work vehicle)
 2: Arm
 3: Bucket
 4: Arm cylinder (hydraulic cylinder)
 5: Bucket cylinder (hydraulic cylinder)
 6: Work device
 8: Vehicle body
 11: Travel device
 20: Engine
 30A: Hydraulic pump
 40: Power generation motor
 43: Travel motor (electrically driven motor)
 57: Mode switching switch (mode switch operation section)
 100, 300, 400, 500: Main controller (controller)

The invention claimed is:

1. A work vehicle comprising:

- an engine that is mounted on a vehicle body;
- a hydraulic pump that is driven by the engine;
- hydraulic cylinders that are extended and contracted by pressurized fluid delivered from the hydraulic pump;
- a work device that is moved according to the extension/contraction operations of the hydraulic cylinders;
- a travel device that is driven independently of the work device;
- an electrically driven motor that is driven by electric power generated by the engine to operate the travel device; and
- a controller that controls the hydraulic cylinders and the electrically driven motor on a basis of a distribution ratio of a first torque and a distribution ratio of a second torque, the first torque being consumed by the work device among torques output by the engine, the second torque being consumed by the travel device among torques output by the engine, wherein

the controller is configured to:

- calculate a reaction force received by the vehicle body when the work vehicle penetrates into an excavation target on a basis of a traction force and a vehicle velocity of the vehicle body;
- determine the distribution ratio of the first torque and the distribution ratio of the second torque such that as the calculated reaction force received by the vehicle body becomes larger, the first torque becomes smaller, the

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second torque becomes larger, and a ratio of an increase in the distribution ratio of the second torque to an increase in the reaction force received by the vehicle body becomes smaller, while the work vehicle is performing excavation work; and
 control output power of the hydraulic pump and output power of the electrically driven motor on the basis of the determined distribution ratio of the first torque and the determined distribution ratio of the second torque.

2. The work vehicle according to claim 1, wherein the controller is configured to:

- decide an upper limit value of the first torque and an upper limit value of the second torque on the basis of the reaction force; and
- control the first torque and the second torque on a basis of the upper limit value of the first torque and the upper limit value of the second torque.

3. The work vehicle according to claim 1, wherein the controller is configured to:

- decide the distribution ratio of the first torque and the distribution ratio of the second torque on the basis of the reaction force after elapse of a predetermined time from penetrating of the work vehicle into the excavation target; and
- control the first torque and the second torque such that the distribution ratio of the first torque and the distribution ratio of the second torque are held until excavation work is finished.

4. The work vehicle according to claim 1, wherein the work device has arms attached to the vehicle body and a bucket attached to the arms, and the controller controls the second torque such that as a pressure of the hydraulic cylinders for operating the arms becomes larger, the second torque becomes smaller.

5. The work vehicle according to claim 1, further comprising:

- a mode switch operation section that can manually switch between a first control mode in which the first torque and the second torque are controlled on the basis of the reaction force and a second control mode in which the first torque and the second torque are controlled such that ratios of the first torque and the second torque are held at predetermined ratios regardless of the reaction force.

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